

### **STRUCTURAL STABILITY: THEORY AND IMPLEMENTATION**

is a practical work that provides engineers and students in structural engineering or structural mechanics with the background needed to make the transition from fundamental theory to practical design rules and computer implementation. Beginning with the basic principles of structural stability and basic governing equations, **STRUCTURAL STABILITY** is a concise and comprehensive introduction that applies the principles and theory of structural stability (which are the basis for structural steel design) to the solution of practical building frame design problems. Special features include:

- modern theories of structural stability of members and frames, and a discussion of how these theories may be utilized to provide design rules and calculation techniques for design
- important governing equations and the classical solutions used in design processes

**Chen  
Lui**

**STRUCTURAL  
THEORY AND IMPLEMENTATION**

# **STRUCTURAL STABILITY**

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# **STRUCTURAL STABILITY**

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**THEORY  
AND  
IMPLEMENTATION**

## PREFACE

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techniques suitable for design office use. In this way, the reader not only will obtain an understanding of the fundamental principles and theory of structural stability from an idealized elastic, perfect system, but also to an inelastic imperfect system that leads to the necessary links between the code rules, design office practice, and the actual structural system in the real world.

The continued rapid development in computer hardware and software in recent years has made it possible for engineers and designers to predict structural behavior quite accurately. The advancement in structural analysis techniques coupled with the increased understanding of structural behavior has made it possible for engineers to adopt the Limit States Design philosophy. A limit state is defined as a condition at which a structural system or its component ceases to perform its intended function under normal conditions (Serviceability Limit State) or failure under severe conditions (Ultimate Limit State). The recently published Load and Resistance Factor Design (LRFD) Specification for the

Professor H. L. Michael of Purdue University for continuing support over many years, and to the graduate students, C. Cheng, L. Duan, and F. H. Wu, among others, for preparing the Answers to Some Selected Problems during their course work on Structural Stability in the spring semester of 1986 in the School of Civil Engineering at Purdue University.

*December, 1986*  
*West Lafayette, IN*

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E.M. Lui

## NOTATION

|                  |   |                                                                                                                            |
|------------------|---|----------------------------------------------------------------------------------------------------------------------------|
| $M_{\text{ext}}$ | = | moment at a section due to externally applied loads                                                                        |
| $M_{\text{int}}$ | = | internal resisting moment of the section                                                                                   |
| $M_{\text{tr}}$  | = | transition moment (in Plastic Design)                                                                                      |
| $M_n$            | = | nominal flexural strength                                                                                                  |
| $M_p$            | = | $ZF_y$ = plastic moment                                                                                                    |
| $M_{\text{px}}$  | = | $1.18M_{\text{px}} \left[ 1 - \left( \frac{P}{P_y} \right) \right] \leq M_{\text{px}}$<br>for H-section about strong axis. |
|                  | = | plastic moment capacity about the strong axis considering the influence of axial load                                      |
| $M_{\text{py}}$  | = | $1.19M_{\text{py}} \left[ 1 - \left( \frac{P}{P_y} \right)^2 \right] \leq M_{\text{py}}$<br>for H-section about weak axis. |
|                  | = | plastic moment capacity about the weak axis considering the influence of axial load                                        |

|                  |   |                                                                    |
|------------------|---|--------------------------------------------------------------------|
| $U_w$            | = | $\frac{1}{2} \int_0^L EC_w \left( \frac{d^2 y}{dz^2} \right)^2 dz$ |
|                  | = | strain energy due to warping restraint torsion                     |
| $V$              | = | $-W_{\text{ext}}$ = potential energy                               |
| $W_{\text{int}}$ | = | $-U$ = work done by the internal resisting forces                  |
| $W_{\text{ext}}$ | = | $-V$ = work done by the external applied forces                    |
| $\Pi$            | = | $U + V$ = total potential energy                                   |

### GEOMETRY AND DIMENSIONS

|       |   |                                     |
|-------|---|-------------------------------------|
| $A$   | = | cross sectional area                |
| $b_t$ | = | flange width                        |
| $C_w$ | = | warping constant                    |
|       | = | $\frac{1}{2} I_y h^2$ for I section |
| $d$   | = | depth                               |

|                  |   |                                                       |
|------------------|---|-------------------------------------------------------|
| $E_{\text{eff}}$ | = | effective modulus                                     |
| $E_r$            | = | reduced modulus                                       |
| $E_t$            | = | tangent modulus                                       |
| $F_y, \sigma_y$  | = | yield stress                                          |
| $G$              | = | shear modulus                                         |
|                  | = | $\frac{E}{2(1 + \nu)} = 11,200 \text{ ksi for steel}$ |
| $\nu$            | = | Poisson's ratio                                       |
|                  | = | 0.3 for steel                                         |

**STABILITY RELATED FACTORS**

|       |   |                                                                   |
|-------|---|-------------------------------------------------------------------|
| $A_F$ | = | amplification factor                                              |
| $B_1$ | = | $P - \delta$ moment amplification factor for beam-columns in LRFD |

|          |   |                                                              |
|----------|---|--------------------------------------------------------------|
|          | = | $1 + \psi \frac{P}{P_{ek}} = \text{effective length factor}$ |
| $\kappa$ | = | $\sqrt{\frac{P_c}{P_{ek}}} = \text{effective length factor}$ |
| $r_i$    | = | load factors                                                 |
| $\phi$   | = | resistance factor                                            |
| $\phi_b$ | = | resistance factor for flexure = 0.90                         |
| $\phi_c$ | = | resistance factor for compression = 0.85                     |

Chapter 1

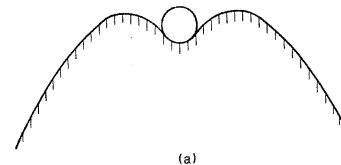
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**GENERAL PRINCIPLES**

change in geometry of the structure must be taken into account; as a consequence, equilibrium equations must be written based on the geometry of a structure that becomes deformed under load. This is known as the *second-order analysis*. The second-order analysis is further complicated by the fact that the resulting equilibrium equations are differential equations instead of the usual algebraic equations. Consequently, a mastery of differential calculus is a must before any attempt to solve these equations.

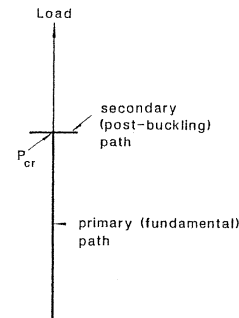
In what follows, we will explain the nature of structural stability and ways to analyze it accurately.

The concept of stability is best illustrated by the well-known example of a ball on a curved surface (Fig. 1.1). For a ball initially in equilibrium, a slight disturbing force applied to the ball on a concave surface (Fig. 1.1a) will displace the ball by a small amount, but the ball will return to its initial equilibrium position once it is no longer being disturbed. In this case, the ball is said to be in a *stable equilibrium*. If the disturbing force is

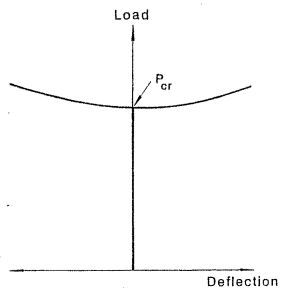


equilibrium or from a state of stable to unstable equilibrium is marked by the *stability limit point*. The tangent stiffness of the system vanishes at just this point. We shall use the principle of vanishing tangent stiffness to calculate the buckling load of a system in subsequent sections and chapters.

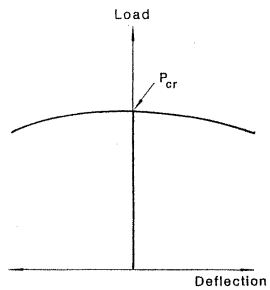
Stability of an elastic system can also be interpreted by means of the concept of *minimum total potential energy*. In nature, an elastic system always tends to go to a state in which the total potential energy is at a minimum. The system is in a stable equilibrium if any deviation from its initial equilibrium state will result in an increase in the total potential energy of the system. The system is in an unstable equilibrium if any deviation from its initial equilibrium state will result in a decrease in total potential energy. Finally, the system is in a neutral equilibrium if any deviation from its initial equilibrium state will produce neither an increase nor a decrease in its total potential energy. Because of this principle, the energy concept can be used to find the buckling load of a system.



## General Principles

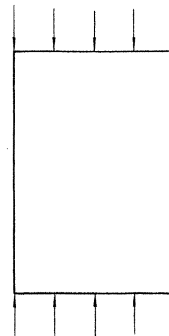


(a) STABLE SYMMETRIC BIFURCATION

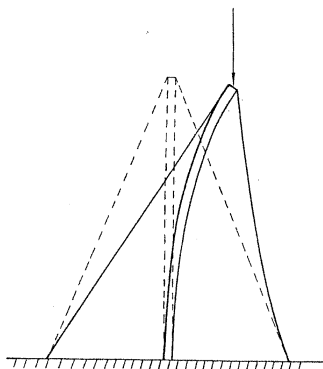


(b) UNSTABLE SYMMETRIC BIFURCATION

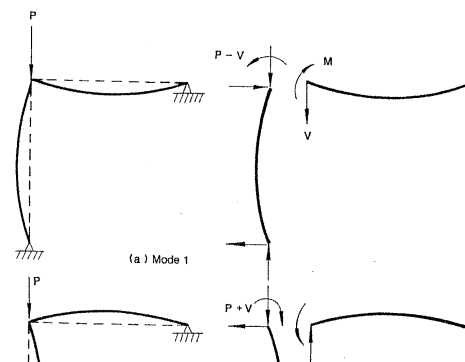
## 1.2 Types of Stability



## General Principles



## 1.2 Types of Stability



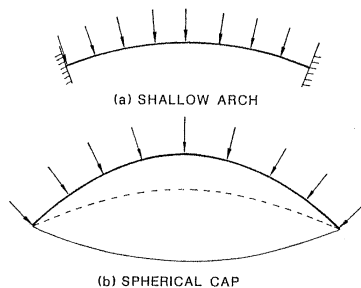


FIGURE 1.8 Examples of limit point buckling

### 1.3 METHODS OF ANALYSES IN STABILITY

The concept of stability as described in Section 1.1 can be used to determine the critical conditions of an elastic system that is susceptible to instability.

#### ***Bifurcation Approach***

The first approach is called the *bifurcation approach*. In this approach, the state at which two or more different but adjacent equilibrium configurations can exist is sought by an *eigenvalue analysis*. The lowest load that corresponds to this state is the critical load of the system. At the critical load, equilibrium can be maintained with alternative deflection modes that are infinitesimally close to one another.

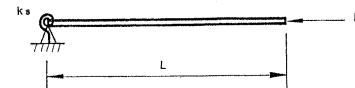
To determine the critical load using the bifurcation approach, it is necessary to identify all possible equilibrium configurations the system can assume at the bifurcation load. This can best be accomplished by

term “conservative forces” used here are those forces whose potential energy is dependent only on the final values of deflection, not the specific paths to reach these final values. If the system is in equilibrium, its total potential energy must be stationary. Thus, by setting the first derivative of the total potential energy function with respect to each generalized displacement equal to zero, we can identify the equilibrium conditions of the system. The critical load can then be calculated from the equilibrium equations.

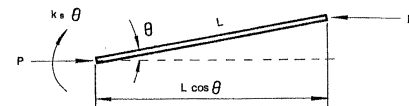
Please note that by setting the first derivative of the total potential energy function equal to zero, we can only identify the equilibrium conditions of the system. To determine whether the equilibrium is stable or unstable, we must investigate higher order derivatives of the total potential energy function.

### Dynamic Approach

The critical load of an elastic system can also be obtained by the *dynamic*



(a)



value of  $k_s/L$ , the system will buckle. At this critical load, equilibrium for the bar is possible in both the original horizontal and slightly deflected positions.

### Energy Approach

In using the energy approach to determine  $P_{cr}$ , one must write an expression for the total potential energy comprising the strain energy and the potential energy of the system. The strain energy stored in the system as the bar assumes its slightly deflected configuration is equal to the strain energy of the spring:

$$U = \frac{1}{2}k_s\theta^2 \quad (1.4.3)$$

The potential energy of the system is the potential energy of the external force and it is equal to the negative of the work done by the external force on the system:

determine not only the critical load,  $P_{cr}$ , but also the nature of the equilibrium of this system. That is, we can further determine whether the equilibrium of the system is stable or unstable at various stages of loading.

To determine whether the equilibrium is stable or unstable for the system in its original ( $\theta = 0$ ) position, we need to investigate the positiveness or negativeness of higher derivatives of the total potential energy function. For this problem, if we take the second derivative of the total potential energy function given in Eq. (1.4.5), we have

$$\frac{d^2\Pi}{d\theta^2} = k_s - PL \cos \theta \approx k_s - PL \quad (1.4.9)$$

Thus, for  $P < P_{cr}$ ,  $d^2\Pi/d\theta^2$  is positive. This indicates that the equilibrium is *stable*. For  $P > P_{cr}$ ,  $d^2\Pi/d\theta^2$  is negative, and the equilibrium condition is therefore *unstable*. This behavior is illustrated in Fig. 1.11, in which the

small  $\theta$  analysis and so no conclusion can be drawn. However, as will be demonstrated in the next section, the stability condition at the critical load ( $P = P_{cr}$ ) can be investigated using the energy approach with a large  $\theta$  analysis.

#### 1.4.2 Rigid Bar Supported by a Translational Spring

Figure 1.12a shows a rigid bar hinged at one end and supported by a linear translational spring of stiffness  $k_s$  at the free end. The spring is assumed to be able to move freely in the horizontal direction but retains its vertical orientation as the bar deflects. The bar is subjected to a concentrated force  $P$  at the free end. Assuming the system is geometrically perfect, we shall determine the critical load of the system.

##### Bifurcation Approach

In Fig. 1.12b we see a slightly deflected position of the bar. Consideration

nontrivial solution, we must have

$$P = P_{cr} = k_s L \quad (1.4.11)$$

##### Energy Approach

The strain energy of the system at its deflected state is equal to the strain energy stored in the structural spring.

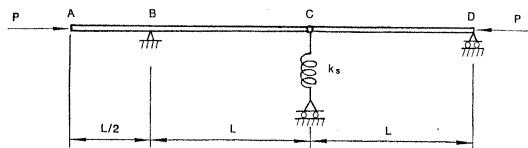
$$U = \frac{1}{2} k_s (L\theta)^2 \quad (1.4.12)$$

The potential energy of the system is equal to the negative of the work done by the applied force.

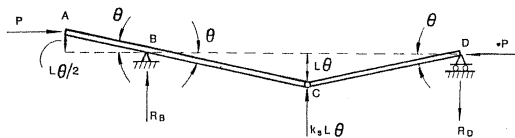
$$V = -PL(1 - \cos \theta) \quad (1.4.13)$$

The total potential energy is

$$\Pi = U + V = \frac{1}{2} k_s (L\theta)^2 - PL(1 - \cos \theta) \quad (1.4.14)$$



(a)



Summing moment about C for the free body in Fig. 1.13c gives

$$R_D L - PL\theta = 0 \quad (1.4.20)$$

Upon substitution of Eq. (1.4.19) into Eq. (1.4.20), we obtain

$$\frac{1}{2}k_s L\theta - \frac{2}{3}P\theta = 0 \quad (1.4.21)$$

The nontrivial solution for the equilibrium equation (1.4.21) gives the critical load of the system as

$$P = P_{cr} = \frac{3}{2}k_s L \quad (1.4.22)$$

### Energy Approach

The strain energy of the system is

$$U = \frac{1}{2}k_s (L\theta)^2 \quad (1.4.23)$$

The potential energy of the system is

#### 1.4.4 Three-Bar System

Figure 1.14a shows a three-bar system supported at the ends A and D by frictionless hinges and connected to one another at B and C by linear rotational springs of spring stiffness  $k_s$ . The system is assumed geometrically perfect in that the springs are unstrained when all the bars are in their horizontal orientation. We shall now determine the critical load  $P_{cr}$  of the system.

##### Bifurcation Approach

In using the bifurcation approach, we are investigating the equilibrium conditions of the system in a slightly deflected configuration. A configuration of the system displaced by an arbitrary kinematic admissible displacement is shown in Fig. 1.14b. A *kinematic admissible displacement* is a displacement that does not violate the constraints of the system. Note that the two parameters ( $\theta_1$  and  $\theta_2$ ) are necessary to describe fully the displaced configuration of the bars. The equilibrium

#### 1.4 Illustrative Examples—Small Deflection Analysis

A and D are zeros. By using the free body diagrams of the left and right bars, respectively (Fig. 1.14c), we can write the following equilibrium equations:

$$k_s(2\theta_1 - \theta_2) - P(L\theta_1) = 0, \quad (1.4.30a)$$

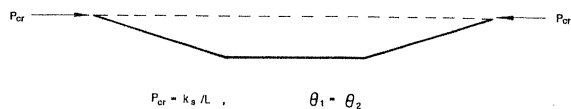
$$k_s(2\theta_2 - \theta_1) - P(L\theta_2) = 0. \quad (1.4.30b)$$

In matrix form, we have

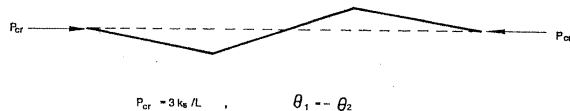
$$\begin{bmatrix} 2k_s - PL & -k_s \\ -k_s & 2k_s - PL \end{bmatrix} \begin{pmatrix} \theta_1 \\ \theta_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}. \quad (1.4.31)$$

For a nontrivial solution, we must have the determinant of the coefficient matrix equal to zero

$$\det \begin{vmatrix} 2k_s - PL & -k_s \\ -k_s & 2k_s - PL \end{vmatrix} = (2k_s - PL)^2 - k_s^2 = 0 \quad (1.4.32)$$



(a) Symmetric Buckling Mode



$$\begin{aligned} \frac{\partial \Pi}{\partial \theta_2} &= -k_s(2\theta_1 - \theta_2) + 2k_s(2\theta_2 - \theta_1) \\ &\quad - PL[\sin \theta_2 - \sin(\theta_1 - \theta_2)] = 0 \end{aligned} \quad (1.4.39b)$$

Upon simplification and using small angle approximation, we can write Eqs. (1.4.39a and b) in matrix form as

$$\begin{bmatrix} 5k_s - 2PL & -4k_s + PL \\ -4k_s + PL & 5k_s - 2PL \end{bmatrix} \begin{pmatrix} \theta_1 \\ \theta_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \quad (1.4.40)$$

For nontrivial solution, we must have

$$\det \begin{vmatrix} 5k_s - 2PL & -4k_s + PL \\ -4k_s + PL & 5k_s - 2PL \end{vmatrix} = 0 \quad (1.4.41)$$

from which the two eigenvalues are

The equilibrium is stable if all of the following conditions are satisfied:

$$\frac{\partial^2 \Pi}{\partial \theta_1^2} > 0 \quad (1.4.45a)$$

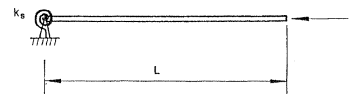
$$\frac{\partial^2 \Pi}{\partial \theta_2^2} > 0 \quad (1.4.45b)$$

$$\left( \frac{\partial^2 \Pi}{\partial \theta_1^2} \right) \left( \frac{\partial^2 \Pi}{\partial \theta_2^2} \right) > \left( \frac{\partial^2 \Pi}{\partial \theta_1 \partial \theta_2} \right)^2 \quad (1.4.45c)$$

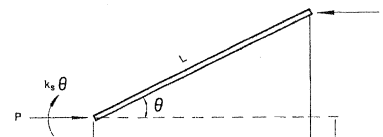
In view of Eqs. (1.4.44a–c), Equations (1.4.45a–c) become

$$P < \frac{5}{2} \left( \frac{k_s}{L} \right) \quad (1.4.46a)$$

$$P < \frac{5}{2} \left( \frac{k_s}{L} \right) \quad (1.4.46b)$$



(a)



The strain energy of the system is equal to the strain energy of the spring

$$U = \frac{1}{2}k_s\theta^2 \quad (1.5.1)$$

The potential energy of the system is the potential energy of the external force

$$V = -PL(1 - \cos \theta) \quad (1.5.2)$$

The total potential energy of the system is then

$$\Pi = U + V = \frac{1}{2}k_s\theta^2 - PL(1 - \cos \theta) \quad (1.5.3)$$

For equilibrium, we must have

$$\frac{d\Pi}{d\theta} = k_s\theta - PL \sin \theta = 0 \quad (1.5.4)$$

This equilibrium equation is satisfied for all values of  $P$  if  $\theta = 0$ . This

postbuckling path is given by

$$P = \frac{k_s\theta}{L \sin \theta} \quad (1.5.5)$$

This path is also plotted in Fig. 1.17 for the range  $-\pi/2 \leq \theta \leq \pi/2$ . It intersects the fundamental path at  $P_{cr} = k_s/L$ .

To study the stability of the equilibrium paths, we need to examine the higher order derivatives of the total potential energy function. By taking the second derivative of  $\Pi$ , we have

$$\frac{d^2\Pi}{d\theta^2} = k_s - PL \cos \theta \quad (1.5.6)$$

For the fundamental path,  $\theta = 0$ , therefore Eq. (1.5.6) becomes

$$\frac{d^2\Pi}{d\theta^2} = k_s - PL \quad (1.5.7)$$

thus the first nonzero term of the series is the fifth term

$$\left. \frac{1}{24} \frac{d^4 \Pi}{d\theta^4} \right|_{\theta=0} \theta^4 = \frac{1}{24} PL\theta^4 \quad (1.5.10)$$

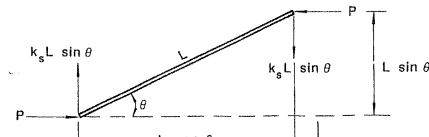
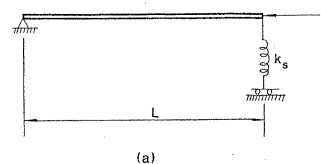
At  $P = P_{cr} = k_s/L$ , we have

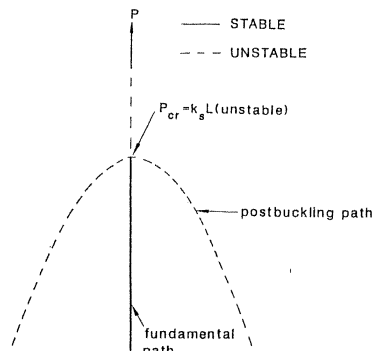
$$\frac{1}{24} PL\theta^4 = \frac{1}{24} k_s \theta^4 \quad (1.5.11)$$

which is positive, indicating that  $\Pi$  is a local minimum and so the equilibrium state at the critical point is stable.

From the above discussion it can be seen that as  $P$  increases gradually to  $P_{cr}$ , the stable fundamental equilibrium path bifurcates into an unstable equilibrium path corresponding to the original horizontal position of the bar and a stable postbuckling equilibrium path corresponding to the deflected configuration of the bar. The equilibrium state at the critical point is stable. The stable postbuckling equilibrium

### 1.5 Illustrative Examples—Large Deflection Analysis





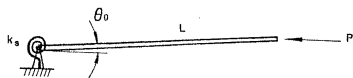
At  $P = P_{cr} = k_s L$ , we have

$$\frac{1}{24}(-4k_s L^2 + PL)\theta^4 = -\frac{1}{8}k_s L^2 \theta^4 \quad (1.5.21)$$

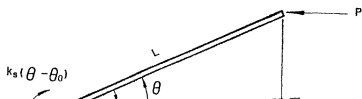
which is negative, indicating that  $\Pi$  is a local maximum and so the equilibrium state at the critical point is unstable.

From the above discussion, it can be seen that as  $P$  increases gradually to  $P_{cr}$ , the stable fundamental equilibrium path bifurcates into an unstable equilibrium path corresponding to the original horizontal position of the bar and an unstable equilibrium path corresponding to the deflected configuration of the bar. The equilibrium state at the critical point is also unstable.

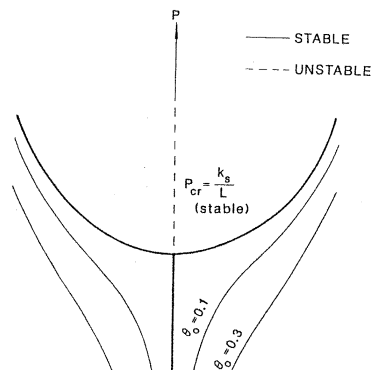
## General Principles



(a)



## 1.6 Illustrative Examples—Imperfect Systems



and they are unstable if

$$P > \frac{k_s}{L \cos \theta} \quad (1.6.8)$$

It is quite obvious that Eq. (1.6.7) always controls for  $-\pi/2 \leq \theta \leq \pi/2$ . Therefore, the equilibrium paths described by Eq. (1.6.5) are always stable, indicating that as  $P$  increases the deflections increase, as shown by the curves with the initial angle  $\theta_0 = 0, 0.1$  and  $0.3$ , and no instability will occur. The maximum load that the system can carry is greater than the critical load,  $P_{cr}$ .

### 1.6.2 Rigid Bar Supported by a Translation Spring

Consider now the imperfect system shown in Fig. 1.22. The system is imperfect, being tilted by an angle  $\theta_0$  when the spring is unstretched. We shall now use the energy approach to study the occurrence of this

#### Energy Approach

The strain energy of the system is equal to the strain energy stored in the spring

$$U = \frac{1}{2} k_s L^2 (\sin \theta - \sin \theta_0)^2 \quad (1.6.9)$$

The potential energy of the system is equal to the potential energy of the external force

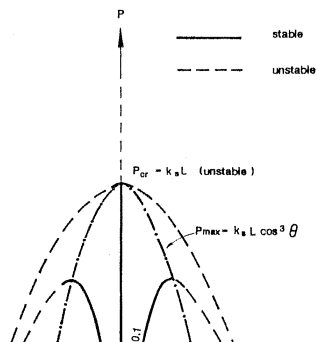
$$V = -PL(\cos \theta_0 - \cos \theta) \quad (1.6.10)$$

The total potential energy of the system is

$$\begin{aligned} \Pi = U + V &= \frac{1}{2} k_s L^2 (\sin \theta - \sin \theta_0)^2 \\ &\quad - PL(\cos \theta_0 - \cos \theta) \end{aligned} \quad (1.6.11)$$

For equilibrium, we must have

$$\frac{d\Pi}{d\theta} = 0$$



and they will be unstable if

$$\sin \theta_0 < \sin^3 \theta \quad (1.6.20)$$

So, it can be seen that the rising equilibrium paths are stable and the falling equilibrium paths are unstable. As the load  $P$  is increased from zero, the load-deflection behavior of the imperfect system will follow the stable rising equilibrium paths as shown in Fig. 1.23, until  $P_{\max}$  is reached, after which the system will become unstable.

From the foregoing examples of imperfect systems, we can conclude that for a system with stable postbuckling equilibrium paths, a small imperfection will not significantly affect the system's behavior. The maximum load the imperfect system can carry is larger than the critical load of the perfect system. On the other hand, for a system with unstable postbuckling equilibrium paths, a small imperfection may have a notice-

where

$R_n$  = nominal resistance of the structural member expressed in unit of stress

$Q_n$  = nominal working, or service stresses computed under working load conditions

$F.S.$  = factor of safety

$i$  = type of load (i.e., dead load, live load, wind load, etc.)

$m$  = number of load type

The left hand side of Eq. (1.7.1) represents the allowable stress of the structural member or component under various loading conditions (for example, tension, compression, bending, shear, etc.). The right hand side of the equation represents the combined stress produced by various load combinations (for example, dead load, live load, wind load, etc.). Formulas for the allowable stresses for various types of structural

Note that in plastic design, safety is incorporated in the load term and is evaluated at the ultimate (plastic strength) limit state.

### 1.7.3 Load and Resistance Factor Design

The purpose of load and resistance factor design (LRFD) is to ensure that the nominal resistance of the structural member or component exceeds that of the load effects. Two safety factors are used, one applied to the loads, and the other to the resistance of the materials. Thus, the load and resistance factor design has the format

$$\phi R_n \geq \sum_{i=1}^m \gamma_i Q_{ni} \quad (1.7.3)$$

where

$R_n$  = nominal resistance of the structural member

$Q_n$  = load effect (e.g., axial force, shear force, bending moment,

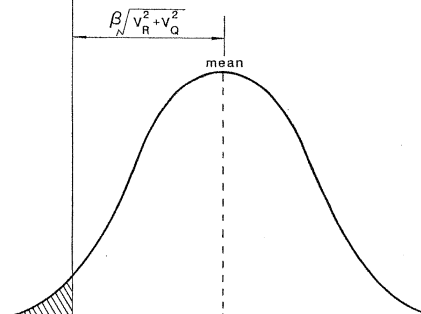
**Table 1.1** Load Factors and Load Combinations<sup>a</sup>

|                                                                             |
|-----------------------------------------------------------------------------|
| 1.4 $D$                                                                     |
| 1.2 $D + 1.6 L + 0.5 (L_r \text{ or } S \text{ or } R)$                     |
| 1.2 $D + 1.6 (L_r \text{ or } S \text{ or } R) + (0.5 L \text{ or } 0.8 W)$ |
| 1.2 $D + 1.3 W + 0.5 L + 0.5 (L_r \text{ or } S \text{ or } R)$             |
| 1.2 $D + 1.5 E + (0.5 L \text{ or } 0.2 S)$                                 |
| 0.9 $D - 1.3 W \text{ or } 1.5 E$                                           |

where

$D$  = dead load  
 $L$  = live load  
 $L_r$  = roof live load  
 $W$  = wind load  
 $S$  = snow load  
 $E$  = earthquake load  
 $R$  = nominal load due to initial rainwater or ice ex-

PROBABILITY  
DENSITY



to stiffness of the structure. One important criterion related to stiffness is that the structure or structural component must not deflect excessively under service load conditions. Thus, regardless of the design method, one should always investigate such *serviceability requirements* as deflection and vibration at service load conditions.

### 1.8 SCOPE

In this book, the discussion of *stability theory* will be limited to conservative systems under static or quasistatic loads. A *conservative system* is a system that is subjected only to conservative forces, that is, to forces whose potential energy is dependent only on the final values of deflection. In addition, most of the material presented in this book's later chapters will be based on what is called *small displacement theory*. Hence, the critical load but not the postbuckling behavior of the member or structure will be studied. Only the stability behavior of structural

### References

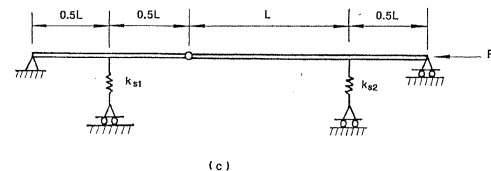
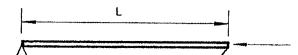


FIGURE P1.1

1.2 Repeat Prob. 1.1, using the energy approach.

1.3 Investigate the stability behavior of the asymmetric spring-bar model shown in Fig. P1.3.

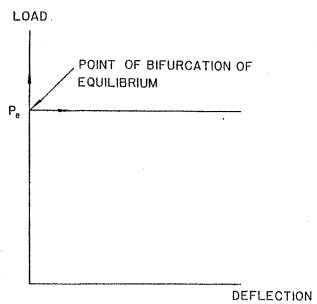
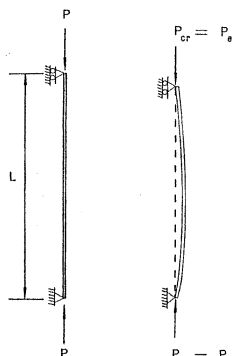


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## Chapter 2

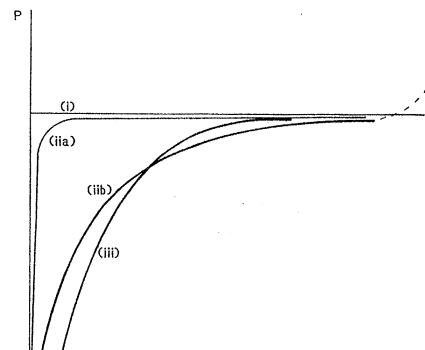
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## COLUMNS



## Columns

## 2.1 Introduction

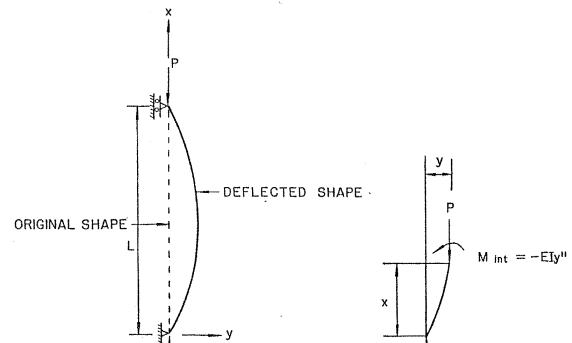


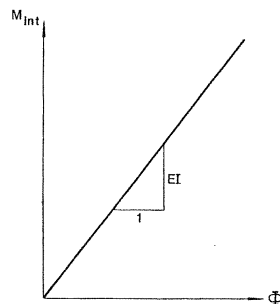
or the *reduced modulus load* ( $P_r$ ). These loads can be obtained by making certain assumptions regarding the strain and stress distributions in the cross section of the column. The effect of this inelasticity can be taken into account by modifying the elastic modulus according to the two inelastic column theories: the *tangent modulus* and the *reduced modulus theories* for a perfectly straight column. These two inelastic column theories will be discussed in Section 2.7.

If the column is not perfectly straight, or if bending exists at the onset of loading, the tangent or reduced modulus theory is not applicable anymore. For such members, the ultimate load  $P_u$  must be determined numerically. Two commonly used numerical procedures to determine  $P_u$  will be discussed in Sections 6.7 and 6.8 of Chapter 6.

## 2.2 CLASSICAL COLUMN THEORY

## 2.2 Classical Column Theory





**FIGURE 2.4** Linear moment-curvature relationship

## 2.2 Classical Column Theory

integration of the moment induced by the stress  $\sigma_x$  over the cross section.

$$M_{\text{int}} = \int_A y_1 \sigma_x dA \quad (2.2.6)$$

Substitution of the kinematic relation (2.2.4) and the stress-strain relation (2.2.5) into the equilibrium equation (2.2.6) gives

$$M_{\text{int}} = \frac{E}{R} \int_A y_1^2 dA \quad (2.2.7)$$

By recognizing that  $\int_A y_1^2 dA$  is the moment of inertia  $I$  of the cross section and  $1/R$  is the curvature  $\Phi$ , Eq. (2.2.7) can easily be reduced to Eq. (2.2.2).

If a small deflection is assumed, the curvature  $\Phi$  can be approximated by the second derivative of the lateral displacement

$$\Phi = \frac{d^2v}{dx^2} \quad (2.2.8)$$

Therefore, we can only determine two unknowns. Substituting the first boundary condition (2.2.14) into Eq. (2.2.13), we have

$$B = 0 \quad (2.2.16)$$

Using the second boundary condition (2.2.15), we have

$$A \sin kL = 0 \quad (2.2.17)$$

Equation (2.2.17) is satisfied if

$$A = 0 \quad (2.2.18)$$

and/or

$$\sin kL = 0 \quad (2.2.19)$$

Equation (2.2.18) is a trivial solution that states that the straight configuration of the column is an equilibrium position. To obtain a

shape and not the amplitude of the deflection can be determined. This is because we approximate the curvature  $\Phi$  to the lateral displacement  $y$  of the column by the linear relation  $\Phi = -y''$  using the small displacement assumption. This approximation leads to the *linear* differential equation [Eq. (2.2.12)]. If the small displacement assumption is obliterated, the resulting differential equation will be nonlinear. The use of formal mathematics for the solution to this nonlinear equation will result in not only the deflected shape but also the amplitude of the deflected column. The discussion of the large displacement behavior of an axially loaded column will be given in the later part of this section.

### 2.2.2 Eccentrically Loaded Column

Figure 2.6a shows the loading condition and the deflected shape of an eccentrically loaded column. The axial force  $P$  is loaded eccentrically at a

using  $k^2 = P/EI$ , we obtain

$$y'' + k^2 y = -k^2 e \quad (2.2.27)$$

Equation (2.2.27) is the governing differential equation of an eccentrically loaded column. The general solution to this equation is

$$y = y_c + y_p \quad (2.2.28)$$

where

$y_c$  = complementary solution to the corresponding homogeneous differential equation (i.e.,  $y'' + k^2 y = 0$ )

$y_p$  = particular solution satisfying Eq. (2.2.27)

The homogeneous solution is given by

$$y_c = A \sin kx + B \cos kx \quad (2.2.29)$$

The particular solution  $y_p$  can be obtained by either the *method of*

The maximum deflection and moment occur at midheight of the column. Therefore, by setting  $x = L/2$ ,  $y_{\max}$  and  $M_{\max}$  are given respectively as

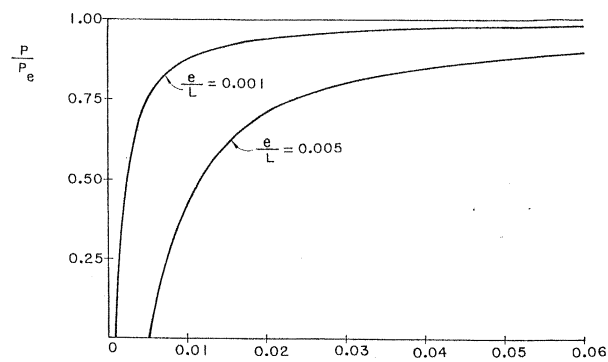
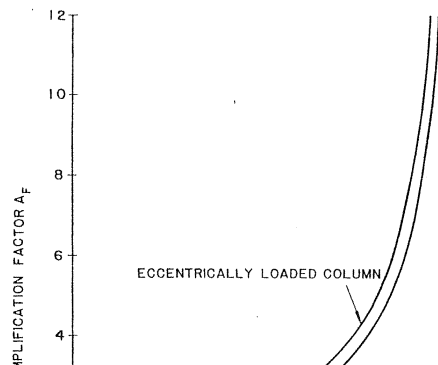
$$y_{\max} = y(L/2) = \left( \frac{1 - \cos kL}{\sin kL} \sin \frac{kL}{2} + \cos \frac{kL}{2} - 1 \right) e \quad (2.2.38)$$

$$M_{\max} = M(L/2) = EI k^2 e \left( \frac{1 - \cos kL}{\sin kL} \sin \frac{kL}{2} + \cos \frac{kL}{2} \right) \quad (2.2.39)$$

Equations (2.2.38) and (2.2.39) can be simplified by using the trigonometric identities

$$\cos kL = 1 - 2 \sin^2 \frac{kL}{2} \quad (2.2.40)$$

$$\sin kL = 2 \sin \frac{kL}{2} \cos \frac{kL}{2} \quad (2.2.41)$$



(2.2.50) as

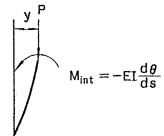
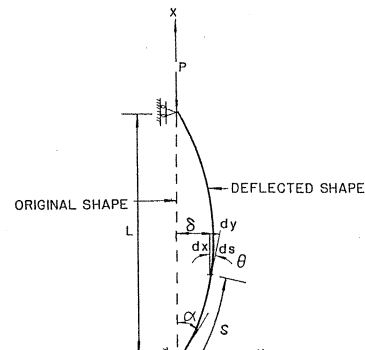
$$\sigma_{\max} = \frac{P}{A} \left[ 1 + \frac{ec}{r^2} \sec \frac{\pi}{2} \sqrt{\frac{P}{P_e}} \right] \quad (2.2.52)$$

This *secant formula* enables us to calculate in a direct manner the maximum stress in an eccentrically loaded column. It will be used in what follows to develop a column strength equation for the design of axially loaded columns with imperfections.

If the first yield of the material is used as the criterion for failure, i.e., if the limit state of the column is defined as the state at which the maximum fiber stress just reaches the yield stress  $\sigma_y$ , the corresponding critical load ( $P_{cr}$ ) can be calculated from Eq. (2.2.52) by setting  $\sigma_{\max} = \sigma_y$ .

$$\sigma_y = \frac{P_{cr}}{A} \left[ 1 + \frac{ec}{r^2} \sec \frac{\pi}{2} \sqrt{\frac{P_{cr}}{P_e}} \right] \quad (2.2.53)$$

## 2.2 Classical Column Theory



we have

$$EI \frac{d^2\theta}{ds^2} + P \sin \theta = 0 \quad (2.2.60)$$

Equation (2.2.60) can be solved by using elliptical integrals. Details of these calculations are given in reference 2. In the following we shall discuss the result of these calculations.

The expression for the midheight deflection of this column as a function of  $P/P_e$  is given as (see reference 2)

$$\frac{\delta}{L} = \frac{2 \sin \frac{\alpha}{2}}{\pi \sqrt{\frac{P}{P_e}}} \quad (2.2.61)$$

where  $\alpha$  is the end slope of the column (Fig. 2.9a).

will be favored; in Fig. 2.10, this stable equilibrium position is represented by curve AC. The deflection modes at various stages of loading on the curve are represented by the inset diagrams. Both the end slope  $\alpha$  and the midheight deflection  $\delta$  increase rapidly as  $P/P_e$  rises slightly above unity. The midheight displacement is the greatest when  $P/P_e$  is about 1.7. After that, further increase on load will result in a decrease in midheight deflection due to the fact that the column has now turned inside out, and the applied force  $P$  will now act as a tensile rather than a compressive force to close the loop.

Some observations and conclusions can be made from the present large displacement analysis:

1. Both the linear and nonlinear theories give the same prediction of the critical load  $(P_{cr}) = P_e$ .
2. When  $P/P_e > 1$ , a slight increase in  $P$  will result in a large increase in displacement.
3. When  $P/P_e$  is about 1.7, the midheight deflection is the greatest.

the buckled shape of the end-restrained member. This will be illustrated in the following examples.

### 2.3.1 Both Ends Fixed

Figure 2.11a shows a column built in at both ends. The forces acting on the column as it buckles are also shown in the figure. From Fig. 2.11b, it can be seen that equilibrium requires that

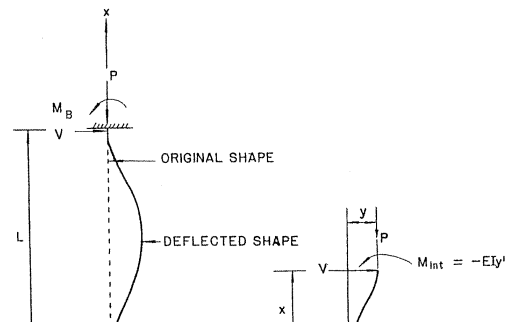
$$-M_{\text{int}} + Py + Vx - M_A = 0 \quad (2.3.1)$$

Since the internal resisting moment is

$$M_{\text{int}} = -EIy'' \quad (2.3.2)$$

The differential equation for equilibrium of this column can be written as

$$EIy'' + L^2y = -\frac{V}{P}x + \frac{M_A}{P} \quad (2.3.3)$$



From Eqs. (2.3.9) and (2.3.10), for a nontrivial solution of  $V$  and  $M_a$ , we must have

$$kL \sin kL + 2 \cos kL - 2 = 0 \quad (2.3.11)$$

By using the trigonometrical identity  $\sin kL = 2 \sin (kL/2) \cos (kL/2)$  and  $\cos kL = 1 - 2 \sin^2 (kL/2)$ , we can write Eq. (2.3.11) as

$$\sin \frac{kL}{2} \left( \frac{kL}{2} \cos \frac{kL}{2} - \sin \frac{kL}{2} \right) = 0 \quad (2.3.12)$$

Equation (2.3.12) can be satisfied if either the first term  $\sin (kL/2)$  or the terms in the parenthesis vanish.

If the first term vanishes, the solution is  $kL = 2n\pi$  where,  $n = 1, 2, 3, \dots$ , from which the critical load is obtained by setting  $n = 1$ , i.e.,

$$P_{cr} = \frac{4\pi^2 EI}{L^2} \quad (2.3.13)$$

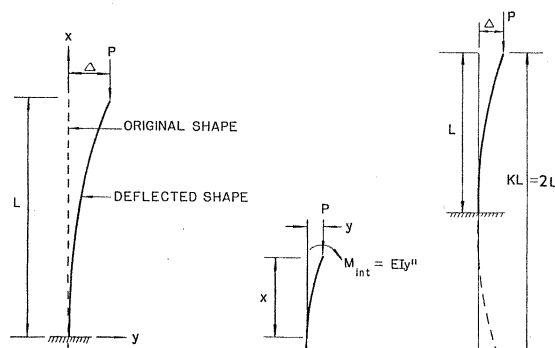
### 2.3 End-Restrained Columns

In other words, the length of the equivalent pinned-pinned column is half that of the fixed-fixed column. Equivalently, the inflection points of the fixed-fixed column are at a distance of  $L/2$  apart (Fig. 2.11c). The factor  $K$  is called the *effective length factor* of the fixed-fixed column. To verify that the inflection points are indeed as shown in Fig. 2.11c, we first write the moment expression along the length of the column and set it equal to zero to calculate the distances  $x$  that give the locations of the inflection points. By differentiating Eq. (2.3.15) twice, we can write the moment expression as

$$M = -EIy'' = EI \frac{M_A}{P} \frac{4\pi^2}{L^2} \cos \frac{2\pi x}{L} = 0 \quad (2.3.18)$$

from which

$$\cos \frac{2\pi x}{L} = 0$$



## 2.3 End-Restrained Columns

which, upon substitution into Eq. (2.3.24), gives

$$y = \Delta(1 - \cos kx) \quad (2.3.29)$$

Using the condition

$$y(L) = \Delta \quad (2.3.30)$$

in Eq. (2.3.29) will give

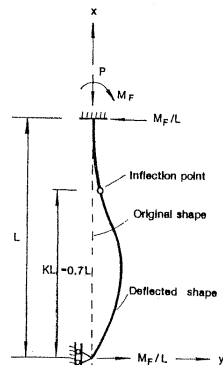
$$\cos kL = 0 \quad (2.3.31)$$

from which

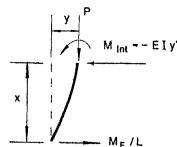
$$kL = \frac{n}{2} \pi, \quad n = 1, 3, \dots \quad (2.3.32)$$

The critical load is the load corresponding to  $n = 1$  or

$$\pi^2 EI \quad (2.3.33)$$



## Columns



## 2.3 End-Restrained Columns

and the deflection function (2.3.39) becomes

$$y = \frac{M_F}{P} \left( \frac{x}{L} - \frac{\sin kx}{\sin kL} \right) \quad (2.3.45)$$

Using the third boundary condition in Eq. (2.3.45) gives

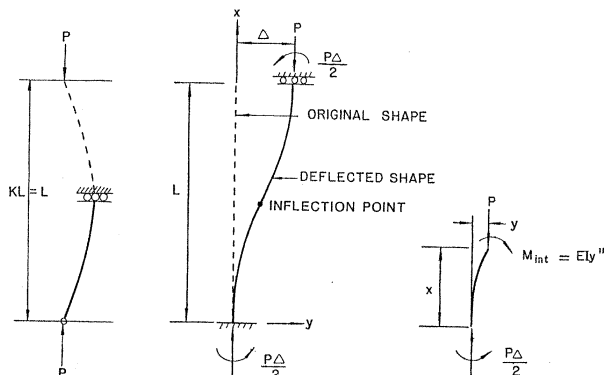
$$\tan kL = kL \quad (2.3.46)$$

from which  $kL$  can be solved by trial and error or by graphical means. The lowest value of  $kL$  that satisfies Eq. (2.3.46) is

$$kL = 4.4934 \quad (2.3.47)$$

which gives

$$P_{cr} = \frac{20.19EI}{L^2} \quad (2.3.48)$$



## 2.3 End-Restrained Columns

Using the boundary condition

$$y(L) = \Delta$$

gives

$$\cos kL = -1 \quad (2.3.58)$$

or

$$kL = n\pi, \quad n = 1, 3, \dots \quad (2.3.59)$$

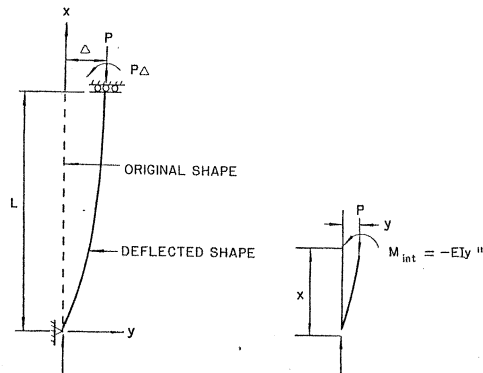
The critical load is given by letting  $n = 1$ , hence

$$P_{cr} = \frac{\pi^2 EI}{L^2} \quad (2.3.60)$$

The deflected shape at  $P_{cr}$  is

$$y = \frac{\Delta}{2} \left( 1 - \cos \frac{\pi x}{L} \right) \quad (2.3.61)$$

and the effective length factor is



### 2.3 End-Restrained Columns

The effective length factor is

$$K = \sqrt{\frac{P_e}{P_{cr}}} = 2 \quad (2.3.77)$$

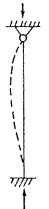
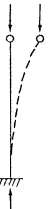
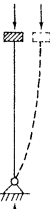
### 2.3.6 AISC Effective Length Factor and Column Curve

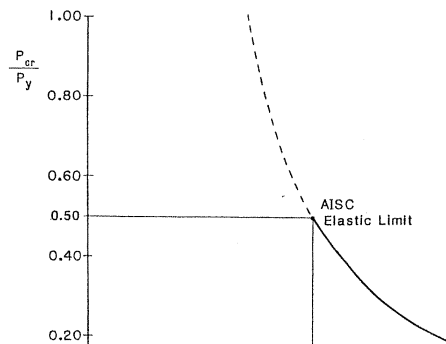
From the above discussion, it can be seen that for an isolated column, regardless of its end conditions, the critical load can be expressed in the general form as

$$P_{cr} = \frac{\pi^2 EI}{(KL)^2} \quad (2.3.78)$$

where  $K$  is the effective length factor that is dependent on the boundary conditions of the column. The theoretical values of  $K$  for various

**Table 2.1** Theoretical and Recommended K Values for Idealized Columns

|                                                 |                                                                                           |                                                                                           |                                                                                           |                                                                                           |                                                                                           |                                                                                           |
|-------------------------------------------------|-------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------|
| Buckled shape of column is shown by dashed line | (a)<br> | (b)<br> | (c)<br> | (d)<br> | (e)<br> | (f)<br> |
| Theoretical K value                             | 0.5                                                                                       | 0.7                                                                                       | 1.0                                                                                       | 1.0                                                                                       | 2.0                                                                                       | 2.0                                                                                       |
| Recommended design value when ideal condi       |                                                                                           |                                                                                           |                                                                                           |                                                                                           |                                                                                           |                                                                                           |



The internal resisting moment is

$$M_{\text{int}} = -EIy'' \quad (2.3.83)$$

Therefore, the differential equation of equilibrium can be written as

$$y'' + k^2 y = \frac{-V}{EI}x + \frac{M_A}{EI} \quad (2.3.84)$$

The general solution of the differential equation is

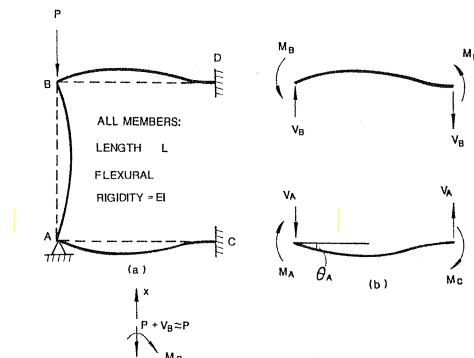
$$y = A \sin kx + B \cos kx - \frac{V}{P}x + \frac{M_A}{P} \quad (2.3.85)$$

Using the displacement and slope boundary conditions for the specific problem and recognizing that

$$M_A = R_{kA} \theta_A \quad (2.3.86)$$

$$M_D = R_{kD} \theta_D \quad (2.3.87)$$

### 2.3 End-Restrained Columns



By referring to the free-body diagram of the lower beam (Fig. 2.18b) and setting  $\theta_c = M_{FA} = M_{FC} = 0$  in the following slope-deflection equations

$$M_A = \frac{EI}{L} (4\theta_A + 2\theta_C) + M_{FA} \quad (2.3.96a)$$

$$M_C = \frac{EI}{L} (4\theta_C + 2\theta_A) + M_{FC} \quad (2.3.96b)$$

we can obtain

$$M_A = \frac{4EI}{L} \theta_A \quad (2.3.97a)$$

$$M_C = \frac{2EI}{L} \theta_A = \frac{1}{2} M_A \quad (2.3.97b)$$

## 2.4 Fourth-Order Differential Equation

falls between  $P_{cr}$  of a pinned-pinned column and that of a fixed-fixed column. The effective length factor for this elastically end-restrained column in the simple frame is

$$K = \sqrt{\frac{P_c}{P_{cr}}} = \sqrt{\frac{\pi^2}{21.03}} = 0.685 \quad (2.3.104)$$

## 2.4 FOURTH-ORDER DIFFERENTIAL EQUATION

In the previous section, the governing differential equation describing the behavior of the column has been developed by consideration of equilibrium for a column segment of *finite* size. The resulting equilibrium equation is second order. Depending on the end conditions, this second-order differential equation may or may not be homogeneous. By enforcing proper geometric (or kinematic) boundary conditions, the critical load can be obtained as the eigenvalue of the characteristic

the *column*. By summing the moment about point b, we obtain

$$Q dx + P dy + M - \left( M + \frac{dM}{dx} dx \right) = 0$$

or, upon simplification

$$Q = \frac{dM}{dx} - P \frac{dy}{dx} \quad (2.4.1)$$

Summing force horizontally, we can write

$$-Q + \left( Q + \frac{dQ}{dx} dx \right) = 0$$

or, upon simplification

$$\frac{dQ}{dx} = 0 \quad (2.4.2)$$

## 2.4 Fourth-Order Differential Equation

### *Pinned-Pinned Column*

For a pinned-pinned column (Fig. 2.20), the four boundary conditions are:

$$y(0) = 0, \quad M(0) = 0 \quad (2.4.8)$$

$$y(L) = 0, \quad M(L) = 0 \quad (2.4.9)$$

Since  $M = -EIy''$ , the moment conditions can be written as

$$y''(0) = 0 \quad (2.4.10)$$

$$y''(L) = 0 \quad (2.4.11)$$

Using the conditions  $y(0) = y''(0) = 0$ , we obtain

$$B = D = 0 \quad (2.4.12)$$

The deflection function (2.4.7) reduces to

In matrix form

$$\begin{bmatrix} \sin kL & L \\ -k^2 \sin kL & 0 \end{bmatrix} \begin{bmatrix} A \\ C \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \quad (2.4.16)$$

If  $A = C = 0$ , the solution is trivial. Therefore, to obtain a nontrivial solution, the determinant of the coefficient matrix of Eq. (2.4.16) must vanish, i.e.,

$$\det \begin{vmatrix} \sin kL & L \\ -k^2 \sin kL & 0 \end{vmatrix} = 0 \quad (2.4.17)$$

or

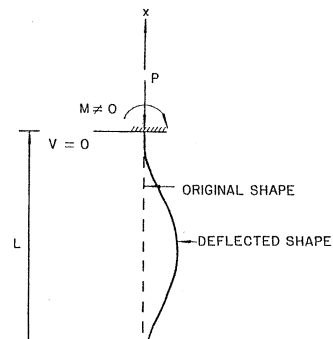
$$k^2 L \sin kL = 0 \quad (2.4.18)$$

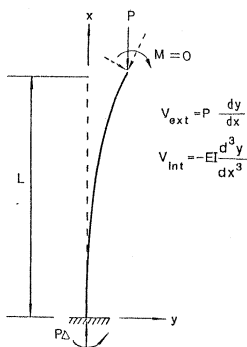
Since  $k^2 \neq 0$ , we must have

$$\sin kL = 0 \quad (2.4.19)$$

or  $kL = n\pi$ ,  $n = 1, 2, 3$

## 2.4 Fourth-Order Differential Equation

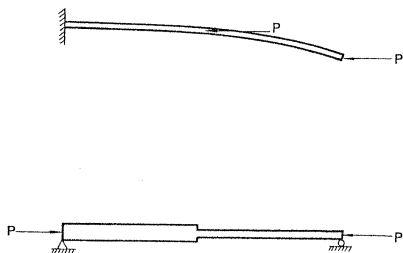




**Table 2.2** Boundary Conditions for Various End Conditions

| End conditions | Boundary conditions |                     |
|----------------|---------------------|---------------------|
| Pinned         | $y = 0,$            | $y'' = 0$           |
| Fixed          | $y = 0,$            | $y' = 0$            |
| Guided         | $y' = 0,$           | $y''' = 0$          |
| Free           | $y'' = 0,$          | $y''' + k^2 y' = 0$ |

For the three cases studied, the solutions of the fourth-order differential equation are seen to lead to the same critical loads as the second-order equation. Note that the second-order equations for these cases studied previously are different because of different boundary conditions, but the fourth-order equation is the same for all cases. In



## 2.5 Special Members

With reference to Fig. 2.24b, the differential equation for segment 1 is

$$EIy_1'' + 2Py_1 = M_A \quad (2.5.1)$$

or

$$y_1'' + k_1^2 y_1 = \frac{M_A}{EI} \quad (2.5.2)$$

where

$$k_1^2 = \frac{2P}{EI} \quad (2.5.3)$$

$$M_A = P(\Delta_B + \Delta_C) \quad (2.5.4)$$

in which  $\Delta_B$  and  $\Delta_C$  are the lateral deflections with respect to the  $x_1 - y_1$  axes for points B and C, respectively. The general solution of Eq. (2.5.2) is

The first two boundary conditions give

$$B = -\frac{M_A}{EIk_1^2} \quad (2.5.15)$$

$$A = 0 \quad (2.5.16)$$

and the continuity conditions (2.5.13) and (2.5.14), give

$$D = -\frac{M_A}{EIk_1^2} \cos \frac{k_1 L}{2} + \frac{M_B}{EIk_1^2} - \frac{M_B}{EIk_2^2} \quad (2.5.17)$$

$$C = \frac{M_A}{EIk_1 k_2} \sin \frac{k_1 L}{2} \quad (2.5.18)$$

Substitution of these expressions into the deflection functions (2.5.5) and (2.5.10) gives

$$v_1 = \frac{M_A}{EIk_1^2} (1 - \cos k_1 x_1) \quad (2.5.19)$$

For a nontrivial solution, we must have

$$\det \begin{vmatrix} k_{11} & k_{12} \\ k_{21} & k_{22} \end{vmatrix} = 0 \quad (2.5.25)$$

By trial and error, and by recognizing that  $k_1 = \sqrt{2} k_2$ , we find that the lowest value of  $k_2$  satisfying Eq. (2.5.25) is

$$k_2 = \frac{1.4378}{L} \quad (2.5.26)$$

Using the definition of  $k_1$  [Eq. (2.5.8)], we obtain

$$P_{cr} = 2.067 \frac{EI}{L^2} \quad (2.5.27)$$

into two segments, 1 and 2, and a differential equation is written for each segment. For convenience, two sets of coordinate axes are again established:  $x_1 - y_1$  for segment 1 and  $x_2 - y_2$  for segment 2.

The fourth-order differential equation for segment 1 is

$$y_1^{IV} + k^2 y_1'' = 0 \quad (2.5.28)$$

where  $k^2 = P/EI$  and the general solution is

$$y_1 = A \sin kx_1 + B \cos kx_1 + Cx_1 + D \quad (2.5.29)$$

Similarly, the fourth-order differential equation for segment 2 is

$$y_2^{IV} + k^2 y_2'' = 0 \quad (2.5.30)$$

and the general solution is

$$y_2 = E \sin kx_2 + F \cos kx_2 + Gx_2 + H \quad (2.5.31)$$

## 2.6 Initially Crooked Columns

(2.5.39) must vanish

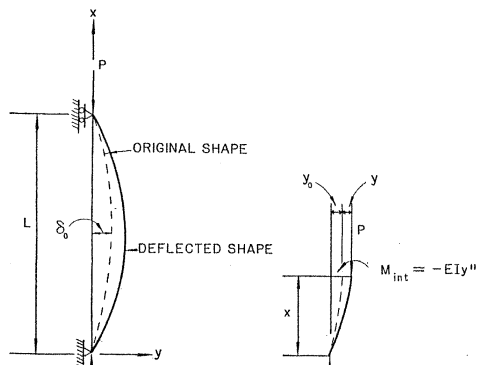
$$\det \begin{vmatrix} \sin \frac{3}{2}kL & \frac{3}{2}L & 0 & 0 \\ 0 & 0 & \sin kL & L \\ k \cos \frac{3}{2}kL & 1 & k \cos kL & 1 \\ -\sin \frac{3}{2}kL & 0 & \sin kL & 0 \end{vmatrix} = 0 \quad (2.5.40)$$

or, after simplification, we obtain the characteristic equation as

$$5 \sin \frac{3}{2}kL \sin kL - 3kL \sin \frac{5}{2}kL = 0 \quad (2.5.41)$$

By trial and error, the smallest value of  $k$  that satisfies the characteristic equation is

$$k = \frac{2.427}{L} \quad (2.5.42)$$



## 2.6 Initially Crooked Columns

The particular solution can be obtained by using the method of undetermined coefficient. Since the right-hand side of the equation consists of a sine and/or cosine term, the particular solution is of the form

$$y_p = C \sin \frac{\pi x}{L} + D \cos \frac{\pi x}{L} \quad (2.6.6)$$

in which  $C$  and  $D$  are the undetermined coefficients.

To determine  $C$  and  $D$ , we substitute Eq. (2.6.6) into the differential equation (2.6.4), and, after combining terms, we obtain

$$\left[ C \left( k^2 - \frac{\pi^2}{L^2} \right) + k^2 \delta_0 \right] \sin \frac{\pi x}{L} + \left[ D \left( k^2 - \frac{\pi^2}{L^2} \right) \right] \cos \frac{\pi x}{L} = 0 \quad (2.6.7)$$

This equation must be satisfied for all values of  $x$ . Thus, the terms in both square brackets must vanish. The vanishing of the first square brackets

from which either

$$A = 0 \quad (2.6.16)$$

or

$$\sin kL = 0 \quad (2.6.17)$$

If we let  $\sin kL = 0$ , we again limit the solution to  $P = P_e$ . Therefore, we must have  $A = 0$ . With  $A = B = 0$ , Eq. (2.6.11) becomes

$$y = \frac{P/P_e}{1 - P/P_e} \delta_0 \sin \frac{\pi x}{L} \quad (2.6.18)$$

Equation (2.6.18) expresses the deflection from the initial crooked position of the column. To obtain the total deflection (i.e., deflection from the  $x$ -axis), we need to add Eq. (2.6.1) to Eq. (2.6.18).

or

$$y_{\text{total}} = y_0 + y$$

## 2.6 Initially Crooked Columns

some general shape, it is advantageous to express this general shape in a Fourier sine series

$$y_0 = \delta_1 \sin \frac{\pi x}{L} + \delta_2 \sin \frac{2\pi x}{L} + \dots \quad (2.6.23)$$

If we proceed as before with each term of the series, and then sum the results together, we have

$$y_{\text{total}} = \frac{\delta_1}{1 - P/P_e} \sin \frac{\pi x}{L} + \frac{\delta_2}{2^2 - P/P_e} \sin \frac{2\pi x}{L} + \dots \quad (2.6.24)$$

### 2.6.2 Perry–Robertson Formula

For an eccentrically loaded column, it has been shown that a formula (the Secant Formula) can be developed that relates the stress in the column to its deflection. For an initially crooked column, a similar approach

$\sigma_{\max} = \sigma_y$ , i.e.,

$$\sigma_y = \frac{P_{cr}}{A} \left[ 1 + \frac{\delta_0 c}{r^2} \frac{1}{1 - P_{cr}/P_e} \right] \quad (2.6.28)$$

The Secant and Perry–Robertson formulas are, strictly speaking, valid only for very long or slender columns for which their slenderness ratios  $L/r$  are large, so that the stress in the column will remain in the elastic range at the buckling load. For shorter columns, material yielding is more important than geometrical imperfection. As a result, it is essential to consider the material imperfection in describing the behavior of shorter or stocky columns. The behavior of inelastic columns will be discussed in the next section.

## 2.7 INELASTIC COLUMNS

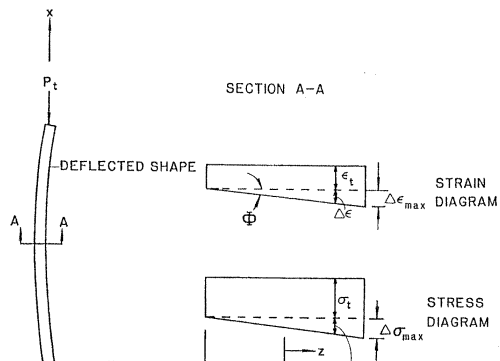
The discussion so far pertains to columns for which the material remains

additional applied force. Since only a portion of the cross section is effective in resisting the axial force at buckling, the elastic modulus  $E$  must be replaced by an *effective modulus*  $E_{eff}$  to describe the behavior of an inelastic column. In this section, we will discuss the buckling behavior of a perfectly straight column buckled in the inelastic range; in particular, we will discuss the *tangent modulus theory* and the *reduced modulus theory* proposed by Engesser<sup>7</sup> and the *inelastic column theory* of Shanley.<sup>8</sup>

### 2.7.1 Tangent Modulus Theory

The *tangent modulus theory* was proposed by Engesser<sup>7</sup> in 1889 to describe the buckling behavior of columns whose buckling stress is above the proportional limit of the material. The following assumptions are made in the tangent modulus theory:

1. The column is perfectly straight.



## 2.7 Inelastic Columns

where  $\sigma$  is the longitudinal stress of a fiber in the cross section and  $z$  is the distance from that fiber to the centroidal axis of the cross section.

From the stress diagram in Fig. 2.28b, it can easily be seen that  $\sigma$  can be expressed as

$$\sigma = \sigma_t + \Delta\sigma = \sigma_t + \frac{\Delta\sigma_{\max}}{h} \left( z + \frac{h}{2} \right) \quad (2.7.3)$$

Substituting Eq. (2.7.3) into Eq. (2.7.2) gives

$$M_{\text{int}} = \int_A \left[ \sigma_t + \frac{\Delta\sigma_{\max}}{h} \left( z + \frac{h}{2} \right) \right] z dA \quad (2.7.4)$$

or, expanding

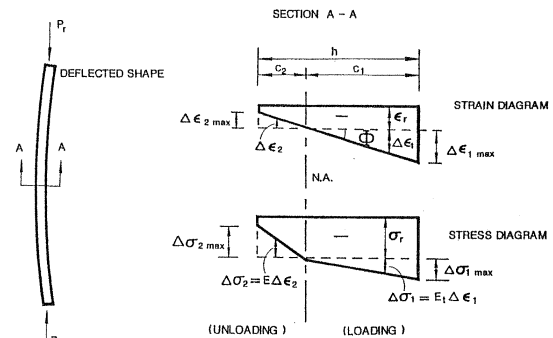
$$M_{\text{int}} = \sigma_t \int_A z dA + \frac{\Delta\sigma_{\max}}{h} \int_A z^2 dA + \frac{\Delta\sigma_{\max}}{2} \int_A z dA \quad (2.7.5)$$

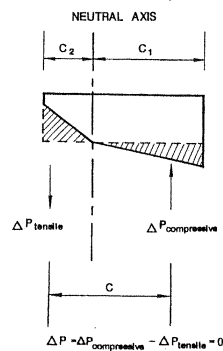
The only difference between Eq. (2.7.11) and Eq. (2.2.10), is that  $E$  in Eq. (2.2.10) is replaced by  $E_t$  in Eq. (2.7.11). As a result, by following the same procedure as in Section 2.2, the critical load of this inelastic column based on the tangent modulus theory is

$$P_t = \frac{\pi^2 E_t I}{L^2} = \frac{E_t}{E} P_e \quad (2.7.12)$$

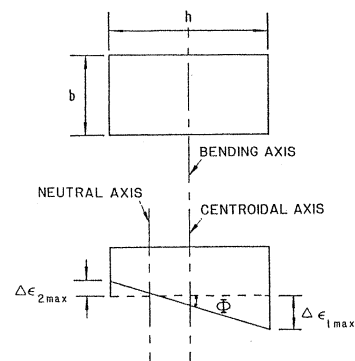
The load expressed in Eq. (2.7.12) is called the *tangent modulus load*. We will show later that this load is the smallest load at which bifurcation of equilibrium of a perfectly straight column can take place in the inelastic range. The *effective modulus* in the tangent modulus theory is therefore the tangent modulus,  $E_t$ . The tangent modulus used in Eq. (2.7.12) depends only on the material property (i.e., stress-strain relationship of the material).

## 2.7 Inelastic Columns





**FIGURE 2.30** Forces due to bending according to the double modulus theory.



we can solve for  $c_1$  and  $c_2$  from Eqs. (2.7.21) and (2.7.22)

$$c_1 = \frac{h\sqrt{E}}{\sqrt{E} + \sqrt{E_t}} \quad (2.7.23)$$

$$c_2 = \frac{h\sqrt{E_t}}{\sqrt{E} + \sqrt{E_t}} \quad (2.7.24)$$

Using Eqs. (2.7.23) and (2.7.19), Eq. (2.7.16a) can be written as

$$\Delta P_{\text{compressive}} = \frac{1}{2}bh^2 \frac{EE_t}{(\sqrt{E} + \sqrt{E_t})^2} \Phi \quad (2.7.25)$$

Similarly, using Eq. (2.7.24) and (2.7.20), Eq. (2.7.16b) can be written as

$$\Delta P_{\text{tensile}} = \frac{1}{2}bh^2 \frac{EE_t}{(\sqrt{E} + \sqrt{E_t})^2} \Phi \quad (2.7.26)$$

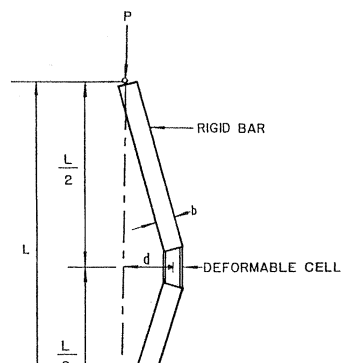
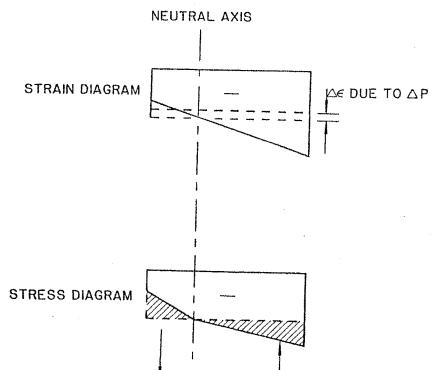
Since the distance,  $c$ , between this pair of forces  $\Delta P_{\text{compressive}}$  and  $\Delta P_{\text{tensile}}$

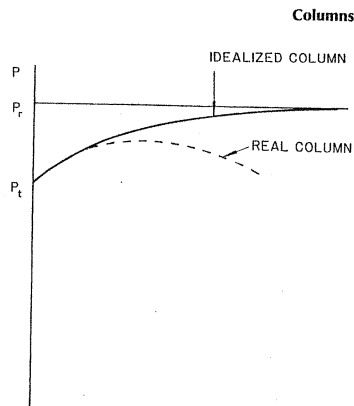
following the same procedure as in Section 2.2, the critical load of the inelastic column is

$$P_r = \frac{\pi^2 E_t I}{L^2} = \frac{E_t}{E} P_e \quad (2.7.34)$$

The load expressed in Eq. (2.7.34) is called the *reduced modulus load*. It is the largest load under which a real column can remain straight. This will be described further in the next section.

In the reduced modulus theory, the effective modulus of the column is of course the reduced modulus,  $E_t$ . Unlike the tangent modulus, the reduced modulus is a function of both material property and geometry of the cross section of a column. In other words, given the same material property, the reduced modulus will be different for different cross-sectional shapes. Thus, the expression for  $E_t$  given in Eq. (2.7.30) is only valid for rectangular cross sections.





**FIGURE 2.34** Postbuckling behavior of column according to Shanley's inelastic analysis.

## 2.8 Design Curves for Aluminum Columns

and recognizing that  $I = Ar^2$ , where  $r$  is the radius of gyration of the cross section, Eq. (2.8.1) can be written as

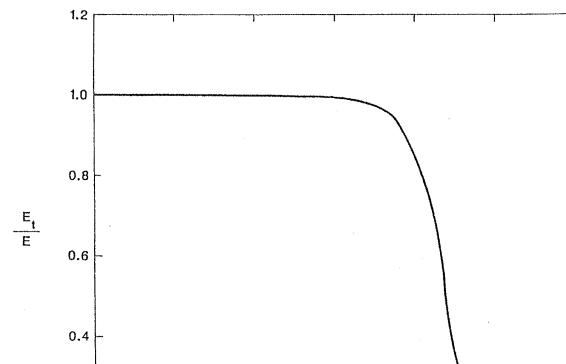
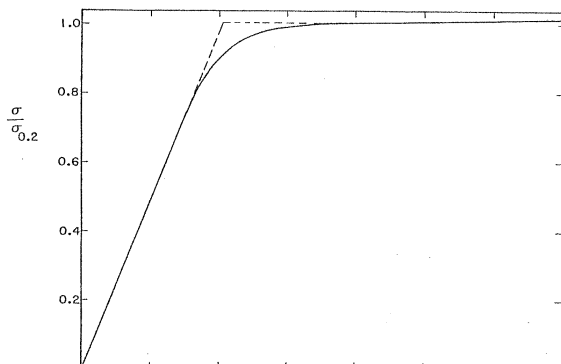
$$\sigma_{cr} = \frac{\pi^2 E_t}{(KL/r)^2} \quad (2.8.3)$$

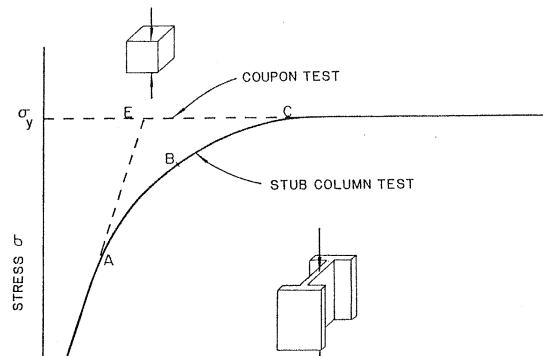
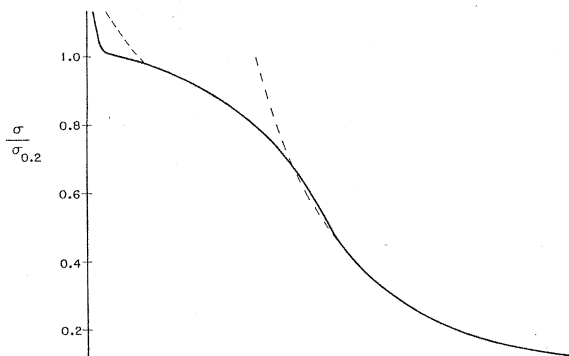
To plot  $\sigma_{cr}$  versus  $KL/r$  using Eq. (2.8.3), we have to know  $E_t$ , which, in turn, is a function of  $\sigma_{cr}$ . Thus, to obtain a proper value of  $E_t$ , we must first know the stress-strain relationship of the material. The stress-strain relationship of aluminum alloys as obtained from experiments (coupon tests) can best be fitted by the Ramberg-Osgood equation.<sup>10</sup>

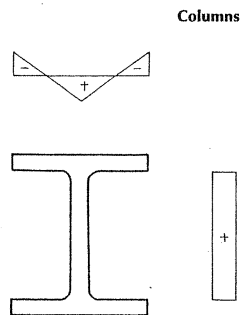
The Ramberg-Osgood equation has the form

$$\varepsilon = \frac{\sigma}{E} + 0.002 \left( \frac{\sigma}{\sigma_{0.2}} \right)^n \quad (2.8.4)$$

where

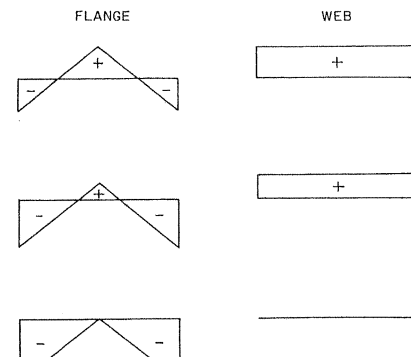






**FIGURE 2.39** Residual stress distribution of a

### 2.9 Stub Column Stress-Strain Curve



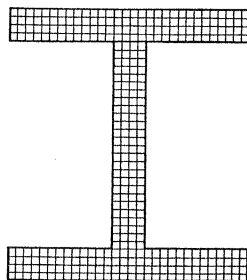


FIGURE 2.41 Discretization of cross section

residual stress is tensile and negative if the residual stress is compressive

## 2.10 Column Curves of Idealized Steel I-Section

7. Evaluate the current state of average stress of the cross section

$$\sigma_{av}^i = \sigma_{av}^{i-1} + \Delta \sigma_{av}^i \quad (2.9.7)$$

8. Finally, evaluate the current state of strain of the cross section from

$$\epsilon^i = \epsilon^{i-1} + \Delta \epsilon^i \quad (2.9.8)$$

Thus, for each value of strain calculated in Step 8, there is a corresponding value of average stress calculated in Step 7. By repeating the process, an average stress versus strain curve can be plotted. This numerically generated  $\sigma - \epsilon$  curve can be used in conjunction with Eq. (2.7.12) to obtain  $P_i$  for the column.  $E_t$  is obtained as the slope of this stress-strain curve.

$E_t$  can also be represented by the expression  $EA_{ei}/A$ . This is demonstrated as follows:

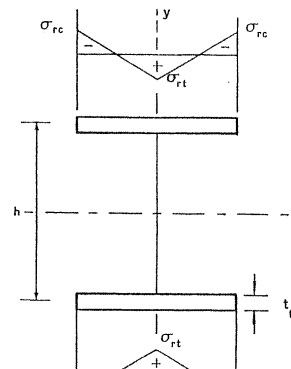
For a given load increment  $\Delta P$ , we can write

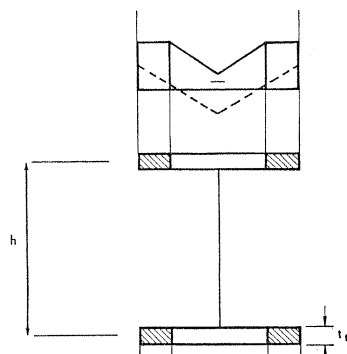
(2.8.4)]. Upon differentiation of this equation, an expression for  $E_t$  can be obtained [Eq. (2.8.5)]. With the tangent modulus known, an aluminum-column curve can be constructed by the procedure outlined in that section.

For steel columns, the stress-strain behavior of a coupon is elastic-perfectly plastic as shown in Fig. 2.38. As a result, the tangent modulus can not be obtained directly from this curve. Instead, it should be obtained from the slope of a stub-column stress-strain curve (Fig. 2.38).

If a stub-column stress-strain curve (obtained either experimentally or numerically) is available, then the critical load of the steel column can be determined directly from the tangent modulus formula [Eq. (2.7.12)], with  $E_t$  determined from the slope of the stub-column stress-strain curve. However, if a stub-column stress-strain curve is not available, an analytical approach can be used to find  $P_{cr}$  for a given value of  $KL/r$  in the inelastic range (for  $KL/r$  in the elastic range, the critical load  $P_{cr}$  is the Euler load,  $P_e$ ). In this approach, an idealized I-section is used. An idealized I-section is an I-section with negligible web thickness. In

## 2.10 Column Curves of Idealized Steel I-Section





Columns

## 2.10 Column Curves of Idealized Steel I-Section

similar triangles, it can be shown then

$$\frac{\sigma_y - \sigma_0}{b_{fe}/2} = \frac{2\sigma_r}{b_f/2} \quad (2.10.8)$$

or

$$\sigma_y - \sigma_0 = 2\sigma_r \frac{b_{fe}}{b_f} \quad (2.10.9)$$

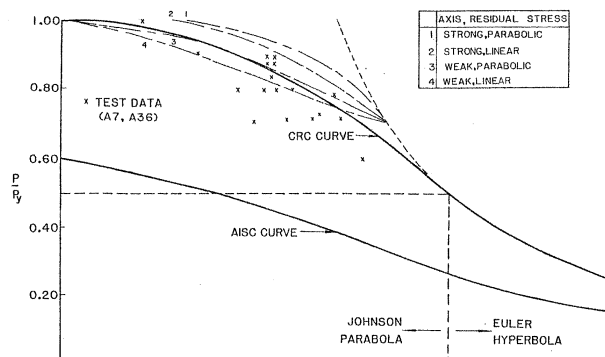
Substituting Eq. (2.10.9) into Eq. (2.10.7) gives

$$P = 2 \left[ \sigma_y b_f t_f - \sigma_r \frac{b_{fe}^2 t_f}{b_f} \right] \quad (2.10.10)$$

or

$$P = A \left[ \sigma_y - \sigma_r \left( \frac{b_{fe}}{b_f} \right)^2 \right] \quad (2.10.11)$$

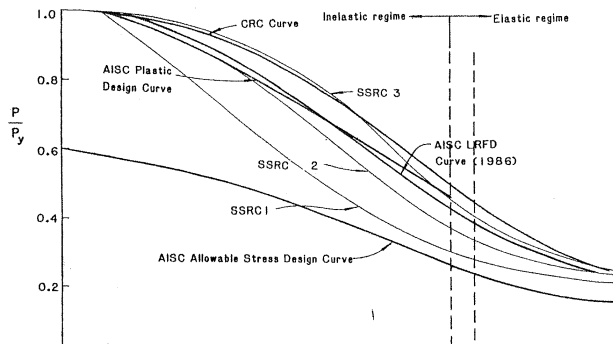
where  $A = 2b_f t_f$  is the area of the cross section of the idealized I section



hot-rolled, wide-flange shapes of mild structural steel, the Column Research Council (CRC) recommended in the first and second editions of the Guide<sup>16</sup> a parabola of the form

$$\sigma_{cr} = \sigma_y - B \left( \frac{KL}{r} \right)^2 \quad (2.11.1)$$

to represent the column strength in the inelastic range. The column strength in the elastic range, however, is represented by the Euler formula. The point of demarcation between inelastic and elastic behavior is chosen to be  $\sigma_{cr} = 0.5\sigma_y$ . The number 0.5 is chosen as a conservative measure of the maximum value of compressive residual stress present in hot-rolled, wide-flange shapes, which is about  $0.3\sigma_y$ . To obtain a smooth transition from the parabola to the Euler curve as well as to maintain a compromise between the strength of columns bent about the strong and weak axes, the constant  $B$  in Eq. (2.11.1) is chosen to be  $\sigma_y^2/4\pi^2E$ . The slenderness ratio that corresponds to  $\sigma_{cr} = 0.5\sigma_y$  is designated as  $C$ .



## 2.11 Design Curves for Steel Columns

format for plastic design of columns is thus

$$\frac{1.7R_n}{F.S.} \geq \gamma \sum_{i=1}^m Q_{ni} \quad (2.11.7)$$

where  $\gamma$  is the *load factor* used in the present AISC-PD Specification. The values for  $\gamma$  are  $\gamma = 1.7$  for live and dead loads only, and  $\gamma = 1.3$  for live and dead loads acting in conjunction with wind or earthquake loads.

### Structural Stability Research Council Curves

Both the ASD curve and PD curve are originated from the CRC curve, which was developed on the basis of the bifurcation concept that assumes the column to be perfectly straight. Although the effect of residual stress is explicitly accounted for, the effect of geometrical imperfections is only accounted for implicitly by applying a variable factor of safety to the basic strength curve.

Curve 2

$$\frac{P}{P_y} = \begin{cases} 1 \text{ (yield level)} & (0 \leq \lambda_c \leq 0.15) \\ 1.035 - 0.202\lambda_c - 0.222\lambda_c^2 & (0.15 \leq \lambda_c \leq 1.0) \\ -0.111 + 0.636\lambda_c^{-1} + 0.087\lambda_c^{-2} & (1.0 \leq \lambda_c \leq 2.0) \\ 0.009 + 0.877\lambda_c^{-2} & (2.0 \leq \lambda_c \leq 3.6) \\ \lambda_c^{-2} \text{ (Euler buckling)} & (\lambda_c \geq 3.6) \end{cases} \quad (2.11.8b)$$

Curve 3

$$\frac{P}{P_y} = \begin{cases} 1 \text{ (yield level)} & (0 \leq \lambda_c \leq 0.15) \\ 1.093 - 0.622\lambda_c & (0.15 \leq \lambda_c \leq 0.8) \\ -0.128 + 0.707\lambda_c^{-1} - 0.102\lambda_c^{-2} & (0.8 \leq \lambda_c \leq 2.2) \\ 0.008 + 0.792\lambda_c^{-2} & (2.2 \leq \lambda_c \leq 5.0) \end{cases} \quad (2.11.8c)$$

Columns may be subjected to later change)

|     |     | steel grade (ASTM) |          |          |          |
|-----|-----|--------------------|----------|----------|----------|
|     |     | Hybrid             |          |          |          |
|     |     | A 514 FL           | A 514 FL | A 441 FL | A 441 FL |
|     |     | 514                | A441 WEB | A36 WEB  | A36 WEB  |
| 1   | 1   | 1                  | 1        | 1        | 1        |
| 2   | 2   | 1                  | 1        | 1        | 1        |
| 3   | 3   | 1                  | 1        | 1        | 1        |
| 4   | 4   | 1                  | 1        | 1        | 1        |
| 5   | 5   | 1                  | 1        | 1        | 1        |
| 6   | 6   | 1                  | 1        | 1        | 1        |
| 7   | 7   | 1                  | 1        | 1        | 1        |
| 8   | 8   | 1                  | 1        | 1        | 1        |
| 9   | 9   | 1                  | 1        | 1        | 1        |
| 10  | 10  | 1                  | 1        | 1        | 1        |
| 11  | 11  | 1                  | 1        | 1        | 1        |
| 12  | 12  | 1                  | 1        | 1        | 1        |
| 13  | 13  | 1                  | 1        | 1        | 1        |
| 14  | 14  | 1                  | 1        | 1        | 1        |
| 15  | 15  | 1                  | 1        | 1        | 1        |
| 16  | 16  | 1                  | 1        | 1        | 1        |
| 17  | 17  | 1                  | 1        | 1        | 1        |
| 18  | 18  | 1                  | 1        | 1        | 1        |
| 19  | 19  | 1                  | 1        | 1        | 1        |
| 20  | 20  | 1                  | 1        | 1        | 1        |
| 21  | 21  | 1                  | 1        | 1        | 1        |
| 22  | 22  | 1                  | 1        | 1        | 1        |
| 23  | 23  | 1                  | 1        | 1        | 1        |
| 24  | 24  | 1                  | 1        | 1        | 1        |
| 25  | 25  | 1                  | 1        | 1        | 1        |
| 26  | 26  | 1                  | 1        | 1        | 1        |
| 27  | 27  | 1                  | 1        | 1        | 1        |
| 28  | 28  | 1                  | 1        | 1        | 1        |
| 29  | 29  | 1                  | 1        | 1        | 1        |
| 30  | 30  | 1                  | 1        | 1        | 1        |
| 31  | 31  | 1                  | 1        | 1        | 1        |
| 32  | 32  | 1                  | 1        | 1        | 1        |
| 33  | 33  | 1                  | 1        | 1        | 1        |
| 34  | 34  | 1                  | 1        | 1        | 1        |
| 35  | 35  | 1                  | 1        | 1        | 1        |
| 36  | 36  | 1                  | 1        | 1        | 1        |
| 37  | 37  | 1                  | 1        | 1        | 1        |
| 38  | 38  | 1                  | 1        | 1        | 1        |
| 39  | 39  | 1                  | 1        | 1        | 1        |
| 40  | 40  | 1                  | 1        | 1        | 1        |
| 41  | 41  | 1                  | 1        | 1        | 1        |
| 42  | 42  | 1                  | 1        | 1        | 1        |
| 43  | 43  | 1                  | 1        | 1        | 1        |
| 44  | 44  | 1                  | 1        | 1        | 1        |
| 45  | 45  | 1                  | 1        | 1        | 1        |
| 46  | 46  | 1                  | 1        | 1        | 1        |
| 47  | 47  | 1                  | 1        | 1        | 1        |
| 48  | 48  | 1                  | 1        | 1        | 1        |
| 49  | 49  | 1                  | 1        | 1        | 1        |
| 50  | 50  | 1                  | 1        | 1        | 1        |
| 51  | 51  | 1                  | 1        | 1        | 1        |
| 52  | 52  | 1                  | 1        | 1        | 1        |
| 53  | 53  | 1                  | 1        | 1        | 1        |
| 54  | 54  | 1                  | 1        | 1        | 1        |
| 55  | 55  | 1                  | 1        | 1        | 1        |
| 56  | 56  | 1                  | 1        | 1        | 1        |
| 57  | 57  | 1                  | 1        | 1        | 1        |
| 58  | 58  | 1                  | 1        | 1        | 1        |
| 59  | 59  | 1                  | 1        | 1        | 1        |
| 60  | 60  | 1                  | 1        | 1        | 1        |
| 61  | 61  | 1                  | 1        | 1        | 1        |
| 62  | 62  | 1                  | 1        | 1        | 1        |
| 63  | 63  | 1                  | 1        | 1        | 1        |
| 64  | 64  | 1                  | 1        | 1        | 1        |
| 65  | 65  | 1                  | 1        | 1        | 1        |
| 66  | 66  | 1                  | 1        | 1        | 1        |
| 67  | 67  | 1                  | 1        | 1        | 1        |
| 68  | 68  | 1                  | 1        | 1        | 1        |
| 69  | 69  | 1                  | 1        | 1        | 1        |
| 70  | 70  | 1                  | 1        | 1        | 1        |
| 71  | 71  | 1                  | 1        | 1        | 1        |
| 72  | 72  | 1                  | 1        | 1        | 1        |
| 73  | 73  | 1                  | 1        | 1        | 1        |
| 74  | 74  | 1                  | 1        | 1        | 1        |
| 75  | 75  | 1                  | 1        | 1        | 1        |
| 76  | 76  | 1                  | 1        | 1        | 1        |
| 77  | 77  | 1                  | 1        | 1        | 1        |
| 78  | 78  | 1                  | 1        | 1        | 1        |
| 79  | 79  | 1                  | 1        | 1        | 1        |
| 80  | 80  | 1                  | 1        | 1        | 1        |
| 81  | 81  | 1                  | 1        | 1        | 1        |
| 82  | 82  | 1                  | 1        | 1        | 1        |
| 83  | 83  | 1                  | 1        | 1        | 1        |
| 84  | 84  | 1                  | 1        | 1        | 1        |
| 85  | 85  | 1                  | 1        | 1        | 1        |
| 86  | 86  | 1                  | 1        | 1        | 1        |
| 87  | 87  | 1                  | 1        | 1        | 1        |
| 88  | 88  | 1                  | 1        | 1        | 1        |
| 89  | 89  | 1                  | 1        | 1        | 1        |
| 90  | 90  | 1                  | 1        | 1        | 1        |
| 91  | 91  | 1                  | 1        | 1        | 1        |
| 92  | 92  | 1                  | 1        | 1        | 1        |
| 93  | 93  | 1                  | 1        | 1        | 1        |
| 94  | 94  | 1                  | 1        | 1        | 1        |
| 95  | 95  | 1                  | 1        | 1        | 1        |
| 96  | 96  | 1                  | 1        | 1        | 1        |
| 97  | 97  | 1                  | 1        | 1        | 1        |
| 98  | 98  | 1                  | 1        | 1        | 1        |
| 99  | 99  | 1                  | 1        | 1        | 1        |
| 100 | 100 | 1                  | 1        | 1        | 1        |

Columns above the curve for the corresponding unstraightened

### AISC Load and Resistance Factor Design Curve

As a result, the AISC Load and Resistance Factor Design (LRFD) Specification<sup>20</sup> adopts the following curve

$$\frac{P}{P_y} = \begin{cases} \exp[-0.419\lambda_c^2] & \lambda_c \leq 1.5 \\ 0.877\lambda_c^{-2} & \lambda_c > 1.5 \end{cases} \quad (2.11.9)$$

to represent column strength. Note that only one curve is recommended for the whole range of possible column strengths. In the development of this curve, the following assumptions were made:

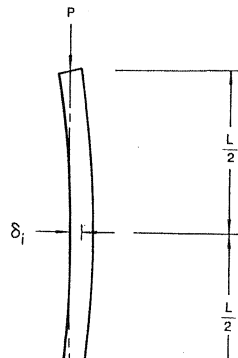
1. The column has small end restraints corresponding to an end-restraint parameter  $G = 10$  (see Chapter 4) or an effective length factor  $K = 0.96$ .
2. The column has an initial crookedness sinusoidal in shape and has an amplitude of  $(1/1500)L$  at midheight.
3. The critical force is calculated with the assumption that the column is perfectly straight.

### 2.11 Design Curves for Steel Columns

**Table 2.4** Summary of Column Curves

| Column curves  | Column equations                                                                                                                                                                                          |
|----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| CRC curve      | $\frac{P}{P_y} = 1 - \frac{\lambda_c^2}{4} \quad \lambda_c \leq \sqrt{2}$                                                                                                                                 |
|                | $\frac{P}{P_y} = \frac{1}{\lambda_c^2} \quad \lambda_c > \sqrt{2}$                                                                                                                                        |
| AISC-ASD curve | $\frac{P}{P_y} = \frac{1 - \frac{\lambda_c^2}{4}}{\frac{5}{3} + \frac{3}{8}\left(\frac{\lambda_c}{\sqrt{2}}\right) - \frac{1}{8}\left(\frac{\lambda_c}{\sqrt{2}}\right)^3} \quad \lambda_c \leq \sqrt{2}$ |
|                | $\frac{P}{P_y} = \frac{12}{23\lambda_c^2} \quad \lambda_c > \sqrt{2}$                                                                                                                                     |
|                | $1.7\left(1 - \frac{\lambda_c^2}{4}\right)$                                                                                                                                                               |

## Columns



## 2.11 Design Curves for Steel Columns

occurs at the most severely stressed fiber, i.e., when

$$\frac{P}{P_y} + \frac{M}{M_y} = 1 \quad (2.11.13)$$

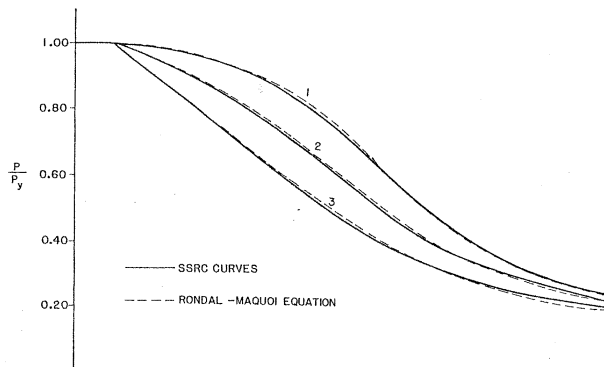
Using Eq. (2.11.12), we have

$$\frac{P}{P_y} + \frac{P\delta_i}{\left(1 - \frac{P}{P_e}\right)M_y} = 1 \quad (2.11.14)$$

or

$$\frac{P}{P_y} + \frac{P\delta_i}{\left(1 - \frac{P}{P_e}\right)M_y} = 1 \quad (2.11.15)$$

Substituting  $M_y = S\sigma_y$  and  $P_y/P_e = \lambda_c^2$  into Eq. (2.11.15) and defining



### 2.11 Design Curves for Steel Columns

Thus, the criterion of failure for the column is

$$\frac{P}{P_t} + \frac{M}{M_m} = 1 \quad (2.11.22)$$

Substituting Eq. (2.11.12) into Eq. (2.11.22) yields

$$\frac{P}{P_t} + \frac{P \delta_1}{\left(1 - \frac{P}{P_c}\right) M_m} = 1 \quad (2.11.23)$$

The flow moment can be expressed as the product of the *plastic section modulus*  $Z$  and an *average flow stress*  $\sigma_0$

$$M_m = Z \sigma_0 \quad (2.11.24)$$

where  $\lambda_c = \frac{1}{\pi} \sqrt{\frac{\sigma_y}{E}} \left( \frac{KL}{r} \right)$ , Eq. (2.11.26) can be written as

$$\left( \frac{P}{P_y} \right) \lambda_c^2 \bar{E} + \frac{P \delta_i}{\left( 1 - \frac{P}{P_y} \lambda_c^2 \right) f \bar{\sigma}_y S \sigma_y} = 1 \quad (2.11.31)$$

or

$$\left( \frac{P}{P_y} \right) \lambda_c^2 \bar{E} + \frac{\frac{P}{P_y}}{1 - \frac{P}{P_y} \lambda_c^2} \bar{\eta} = 1 \quad (2.11.32)$$

in which

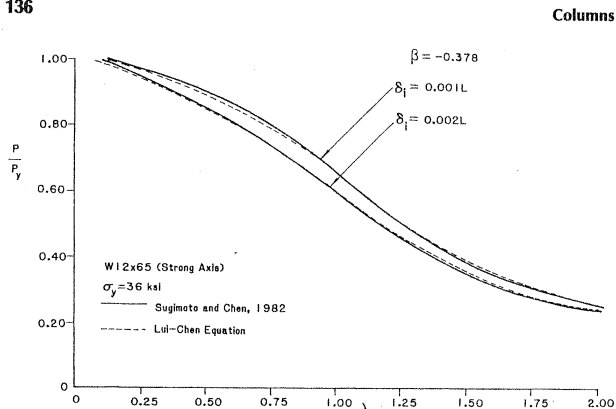
$$\bar{\eta} = \frac{\delta_i A}{f \bar{\sigma}_y S} \quad (2.11.33)$$

or, in terms of  $\lambda_c$  (with  $K = 1$  for pinned-ended column)

$$\bar{\eta} = \pi \sqrt{\frac{E}{\sigma_y}} (\gamma) \left( \frac{1}{f} \right) \left( \frac{c}{r} \right) \left( \frac{\lambda_c}{\bar{\sigma}_y} \right) \quad (2.11.38)$$

Note that this imperfection parameter reflects the effects of yield stress ( $\sigma_y$ ), geometric imperfection ( $\gamma$ ), axis of bending ( $f$ ), cross-sectional shape ( $c, r$ ), and slenderness ratio ( $\lambda_c$ ) on the load-carrying capacity of columns.

The average flow stress  $\sigma_0$  depends on the degree of plastification of the cross section and is a function of the load level and column types. Since for a column the degree of plastification depends on the load level and the load level is a function of the slenderness ratio of the column, this flow stress can be thought of as a function of the slenderness parameter  $\lambda_c$ . If  $\lambda_c$  is very large,  $P/P_y$  is very small, the problem resembles a beam problem and the *plastic limit moment*  $M_{pc}$  will govern the ultimate state, so  $\sigma_0$  will approach  $\sigma_y$ . On the other hand, if  $\lambda_c$  is



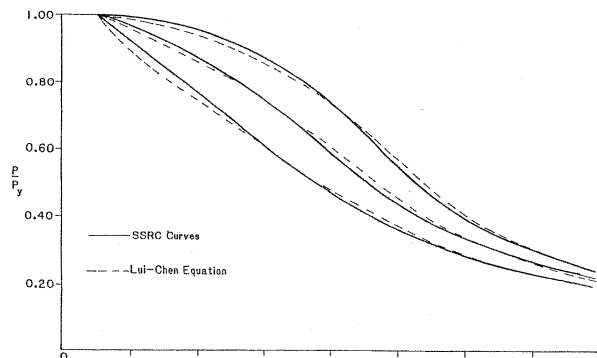
## 2.12 Summary

**Table 2.6** Values of  $\bar{a}$  and  $\bar{b}$  for SSRC Multiple-Column Curves

| SSRC curve | $\bar{a}$ | $\bar{b}$ |
|------------|-----------|-----------|
| 1          | 0.002     | -0.001    |
| 2          | -0.036    | 0.159     |
| 3          | -0.092    | 0.453     |

tions. For small and medium-size hot-rolled, wide-flange shapes, this modulus ratio can be conveniently taken as the ratio of the Euler curve to the CRC curve, i.e.,

$$\bar{E} = \frac{E}{P_c} \sqrt{\frac{4}{(1 + 12\lambda_c^2)}} \quad (\lambda_c \leq \sqrt{2}) \quad (2.11.11)$$



more convenient form that is independent of end conditions can be obtained by differentiating Eq. (2.12.1) twice to give

$$EIy^{IV} + Py'' = 0 \quad (2.12.3)$$

whose general solution is

$$y = A \sin kx + B \cos kx + Cx + D \quad (2.12.4)$$

Alternatively, Eq. (2.12.3) can be obtained by considering equilibrium of an infinitesimal segment of a column. Note that there are five unknowns ( $k$ ,  $A$ ,  $B$ ,  $C$ , and  $D$ ) in Eq. (2.12.4) and there are only four boundary conditions (geometrical and natural). Again, this is an eigenvalue problem. The lowest eigenvalue to Eq. (2.12.4) will thus give the critical load of the column.

For end-restrained columns, it is convenient to modify the unbraced length of the column to an equivalent length of a pinned-ended column so that the column curves prepared for pinned-ended columns can be

is allowed, whereas in the *reduced modulus theory*, a complete strain reversal is allowed, and where a complete strain reversal is assumed to occur on the convex side of the column as it buckles. Although the reduced modulus theory is theoretically correct, the validity of the tangent modulus theory for predicting the buckling strength of real columns is demonstrated and explained by the *Shanley's inelastic column theory*.

Since the tangent modulus load represents a lower bound to the buckling strength of real columns, and since it is easier to evaluate than the reduced modulus load, the tangent modulus theory is used extensively to develop column strength curves for the purpose of design. For instance, the CRC column curve was developed on the basis of the tangent modulus concept. By introducing a safety factor to the CRC curve to account for imperfections of the columns and load eccentricities, one can develop the AISC-ASD curve, which is contained in the present ASD Specification.<sup>3</sup>

## Problems

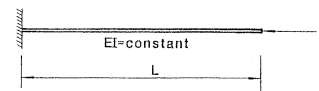
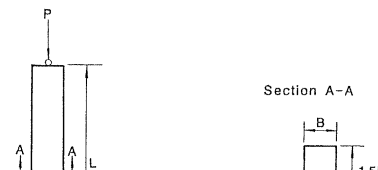


FIGURE P2.1

- 2.2 Find the buckling load of the rectangular section pinned at both ends (Fig. P2.2).



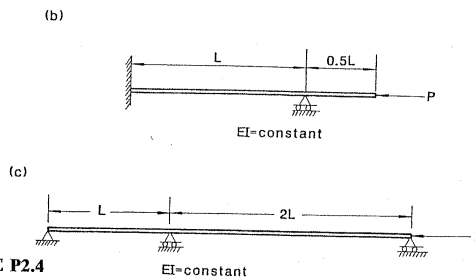


FIGURE P2.4

2.5 For the eccentrically loaded column shown in Fig. P2.5, find an approximate expression for the amplification factor  $A_F$  in the form  $A_F = a/(1 - P/P_e)$  where  $a = f(P/P_e)$ . Compare it with the exact expression  $A_F = \sec[(\pi/\gamma)\sqrt{P/P_e}]$ .

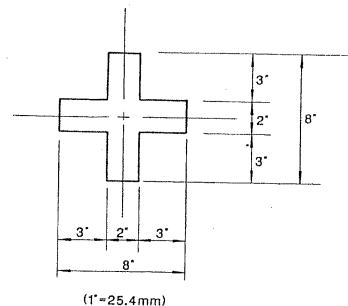
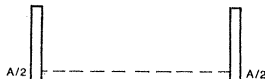


FIGURE P2.8

- 2.11 Design a W-section pinned at both ends and braced at midheight in the weak direction (Fig. P2.11) with an occupancy live load ( $L_o$ ) of 60 kips (267 kN), a roof live load ( $L_r$ ) of 40 kips (178 kN), and a dead load ( $D_o$ ) of 60 kips (267 kN) based on
- the ASD format
  - the PD format
  - the LRFD format
- 2.12 Derive the reduced modulus of elasticity  $E_r$  for an idealized I-section shown in Fig. P2.12, in which it is assumed that one-half of the cross-section area is concentrated in each flange and the area of the web is disregarded.



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#### *Further Reading*

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Chen, W. F., and Atsuta, T. Theory of Beam-Columns. Vol. 1: In-Plane Behavior and Design, 1976; and Vol. 2: Space Behavior and Design. McGraw-Hill, New York, 1977.

## Chapter 3

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# BEAM-COLUMNS

deflection (and moment) caused by the axial-force effect as *secondary deflection* (and moment). Note that the words primary and secondary are used solely for the purpose of convenience and not for the purpose of representing the relative importance of the two effects. In fact, the secondary deflection (and moment) of a beam-column caused by the axial-force effect is sometimes more significant than that caused by the primary bending effect.

Although the analysis of a beam-column is more complicated than that of a beam or a column, closed-form solutions of most beam-columns are available so long as they stay within the realm of purely *elastic* behavior in which the moment can be related to the curvature by a *linear* relationship. (See, for example, *Theory of Elastic Stability* by Timoshenko and Gere.<sup>1</sup>) If yielding or inelasticity occurs in the member, the moment-curvature relationship becomes *nonlinear*. In such cases the use of formal mathematics for the solution of the governing differential equations become intractable and recourse must be had to numerical

### 3.2 BEAM-COLUMN WITH UNIFORMLY DISTRIBUTED LATERAL LOAD

#### 3.2.1 The Closed-Form Solution (Fig. 3.1)

To begin our discussion of the elastic behavior of a beam-column, let us consider a simply supported beam-column subjected to an axial force  $P$  and uniformly distributed lateral load of intensity  $w$  as shown in Fig. 3.1a. A free-body diagram of a segment of the beam-column of length  $x$  from the left support is shown in Fig. 3.1b. The external moment acting on the cut section, is

$$M_{\text{ext}} = Py - \frac{w}{2}x^2 + \frac{wL}{2}x \quad (3.2.1)$$

If elastic behavior is assumed and if the material obeys Hooke's Law, the internal moment  $M_{\text{int}}$  is related to the bending curvature  $y''$  by the linear relationship

or

$$y'' + k^2 y = \frac{w}{2EI} x^2 - \frac{wL}{2EI} x \quad (3.2.4)$$

where

$$k^2 = \frac{P}{EI} \quad (3.2.5)$$

The general solution to Eq. (3.2.4) consists of a complementary solution  $y_c$  and a particular solution  $y_p$ , i.e.,

$$y = y_c + y_p \quad (3.2.6)$$

The complementary solution that satisfies the homogeneous differential equation

$$y'' + k^2 y = 0 \quad (3.2.7)$$

$$C_2 k^2 = -\frac{wL}{2EI} \quad (3.2.15)$$

$$2C_1 + C_3 k^2 = 0 \quad (3.2.16)$$

from which

$$C_1 = \frac{w}{2EI k^2} \quad (3.2.17)$$

$$C_2 = -\frac{wL}{2EI k^2} \quad (3.2.18)$$

$$C_3 = -\frac{2C_1}{k^2} = -\frac{w}{EI k^4} \quad (3.2.19)$$

Hence, the particular solution is

Thus, Eq. (3.2.21) can be written as

$$y = \frac{w}{EI k^4} \left[ \tan \frac{kL}{2} \sin kx + \cos kx - 1 \right] - \frac{w}{2EI k^2} x(L-x) \quad (3.2.26)$$

Introducing the notation

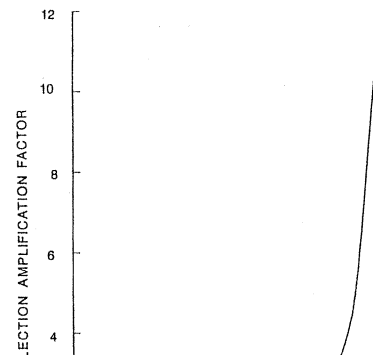
$$u = \frac{kL}{2} \quad (3.2.27)$$

Eq. (3.2.26) can be written as

$$y = \frac{wL^4}{16EIu^4} \left[ \tan u \sin \frac{2ux}{L} + \cos \frac{2ux}{L} - 1 \right] - \frac{wL^2}{8EIu^2} x(L-x) \quad (3.2.28)$$

from which the moment distribution along the length of the member is

### 3.2 Beam-Column with Uniform Distribution Lateral Load



different load combinations can be obtained simply as the sum of the deflections resulting from each individual load application with the same axial load. This is known as the *principle of superposition*. The principle of superposition has been used extensively for the special case of beam problems for which  $P = 0$ . Here, it shows that this principle holds also for beam-column problems for which  $P \neq 0$  provided that (1) the axial force in the general case of the member remains constant, and (2) the same axial force is applied to each of the component cases. We shall use this principle later in the chapter to evaluate the fixed-end moments of beam-columns (Section 3.5).

Although the lateral deflection is directly proportional to the applied lateral load for a constant axial force, this deflection varies nonlinearly with the axial force. This is true even if the lateral load remains unchanged. Also, the proportionality between lateral deflection and lateral load will be destroyed if the axial force varies during the application of the lateral load.

**Table 3.1** Theoretical and Design Deflection Amplification Factors for a Uniformly Loaded Beam-Column

| $u = \frac{kL}{2} = \frac{\pi}{2} \sqrt{\frac{P}{P_e}}$ | Theoretical<br>Eq. (3.2.30) | Design<br>Eq. (3.2.35) |
|---------------------------------------------------------|-----------------------------|------------------------|
| 0                                                       | 1.000                       | 1.000                  |
| 0.20                                                    | 1.016                       | 1.016                  |
| 0.40                                                    | 1.070                       | 1.069                  |
| 0.60                                                    | 1.173                       | 1.171                  |
| 0.80                                                    | 1.354                       | 1.350                  |
| 1.00                                                    | 1.690                       | 1.681                  |
| 1.20                                                    | 2.400                       | 2.402                  |
| 1.40                                                    | 4.822                       | 4.863                  |
| $\pi/2$                                                 | $\infty$                    | $\infty$               |

Using the expression for  $y_{\max}$  [Eq. (3.2.30)] in Eq. (3.2.37), it can easily be shown that Eq. (3.2.36) is obtainable.

For the purpose of design application, we shall again simplify Eq. (3.2.36) by using the power series expansion for  $\sec u$  [Eq. (3.2.31)] in Eq. (3.2.36).

$$M_{\max} = M_0 [1 + 0.4167u^2 + 0.1694u^4 + 0.06870u^6 + \dots] \quad (3.2.38)$$

Using the expression for  $u$  in Eq. (3.2.33), it can be shown that

$$M_{\max} = M_0 \left[ 1 + 1.028 \left( \frac{P}{P_e} \right) + 1.031 \left( \frac{P}{P_e} \right)^2 + 1.032 \left( \frac{P}{P_e} \right)^3 + \dots \right] \quad (3.2.39)$$

or

$$M_{\max} = M_0 \left\{ 1 + \left[ 1.028 \left( \frac{P}{P_e} \right) \right] \left[ 1 + 1.003 \left( \frac{P}{P_e} \right) \right] \right\}$$

### 3.3 Beam-Column with a Concentrated Lateral Load

**Table 3.2** Theoretical and Design Moment Amplification Factors for a Uniformly Loaded Beam-Column

| $u = \frac{kL}{2} = \frac{\pi}{2} \sqrt{\frac{P}{P_e}}$ | Theoretical<br>Eq. (3.2.36) | Design<br>Eq. (3.2.41) |
|---------------------------------------------------------|-----------------------------|------------------------|
| 0                                                       | 1.000                       | 1.000                  |
| 0.20                                                    | 1.017                       | 1.016                  |
| 0.40                                                    | 1.071                       | 1.069                  |
| 0.60                                                    | 1.176                       | 1.171                  |
| 0.80                                                    | 1.360                       | 1.350                  |
| 1.00                                                    | 1.702                       | 1.681                  |
| 1.20                                                    | 2.444                       | 2.402                  |
| 1.40                                                    | 4.983                       | 4.863                  |
| $\pi/2$                                                 | $\infty$                    | $\infty$               |

Rearranging and using the expression  $k^2 = P/EI$ , we obtain

$$y'' + k^2 y = -\frac{Q(L-a)}{LEI} x \quad \text{for } 0 \leq x \leq a \quad (3.3.2a)$$

$$y'' + k^2 y = -\frac{Qa(L-x)}{LEI} \quad \text{for } a \leq x \leq L \quad (3.3.2b)$$

The general solutions are

$$y = A \sin kx + B \cos kx - \frac{Q(L-a)}{LEIk^2} x \quad \text{for } 0 \leq x \leq a \quad (3.3.3a)$$

$$y = C \sin kx + D \cos kx - \frac{Qa(L-x)}{LEIk^2} \quad \text{for } a \leq x \leq L \quad (3.3.3b)$$

from which

$$Q(L-a)$$

and (3.3.3b) gives

$$y = \frac{Q}{EIk^3} \frac{\sin k(L-a)}{\sin kL} \sin kx - \frac{Q(L-a)}{LEIk^2} x \quad \text{for } 0 \leq x \leq a \quad (3.3.13a)$$

$$y = -\frac{Q \sin ka}{EIk^3 \tan kL} \sin kx + \frac{Q \sin ka}{EIk^3} \cos kx - \frac{Qa(L-x)}{LEIk^2} \quad \text{for } a \leq x \leq L \quad (3.3.13b)$$

from which

$$y' = \frac{Q}{EIk^2} \frac{\sin k(L-a)}{\sin kL} \cos kx - \frac{Q(L-a)}{LEIk^2} \quad \text{for } 0 \leq x \leq a \quad (3.3.14a)$$

in brackets in Eqs. (3.3.16) and (3.3.17) are thus the *theoretical deflection and moment amplification factors*.

To simplify the expressions for the maximum deflection equation (3.3.16) and the maximum moment equation (3.3.17), we use the power series expansion for  $\tan u$

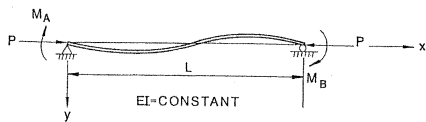
$$\tan u = u + \frac{1}{3}u^3 + \frac{2}{15}u^5 + \frac{17}{315}u^7 + \dots \quad (3.3.18)$$

Upon substituting Eq. (3.3.18) into Eqs. (3.3.16) and (3.3.17) and simplifying, it can be shown that these equations can be written approximately as

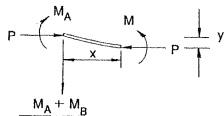
$$y_{\max} \approx y_0 \left[ \frac{1}{1 - \left( \frac{P}{P_e} \right)} \right] \quad (3.3.19)$$

**Table 3.4** Theoretical and Design Moment Amplification Factors for a Beam-Column with a Concentrated Lateral Load at Midspan

| $u = \frac{kL}{2} = \frac{\pi}{2} \sqrt{\frac{P}{P_e}}$ | Theoretical<br>Eq. (3.3.17) | Design<br>Eq. (3.3.20) |
|---------------------------------------------------------|-----------------------------|------------------------|
| 0                                                       | 1.000                       | 1.000                  |
| 0.20                                                    | 1.014                       | 1.013                  |
| 0.40                                                    | 1.057                       | 1.055                  |
| 0.60                                                    | 1.140                       | 1.137                  |
| 0.80                                                    | 1.287                       | 1.280                  |
| 1.00                                                    | 1.557                       | 1.545                  |
| 1.20                                                    | 2.143                       | 2.122                  |
| 1.40                                                    | 4.141                       | 4.090                  |
| $\pi/2$                                                 | $\infty$                    | $\infty$               |



(a)



## 3.4 Beam-Columns Subjected to End Moments

Therefore, Eq. (3.4.4) can now be written as

$$y = -\frac{(M_A \cos kL + M_B)}{EI k^2 \sin kL} \sin kx + \frac{M_A}{EI k^2} \cos kx + \frac{M_A + M_B}{LEI k^2} x - \frac{M_A}{EI k^2} \quad (3.4.8)$$

from which

$$y' = -\frac{(M_A \cos kL + M_B)}{EI k \sin kL} \cos kx - \frac{M_A}{EI k} \sin kx + \frac{M_A + M_B}{LEI k^2} \quad (3.4.9)$$

and

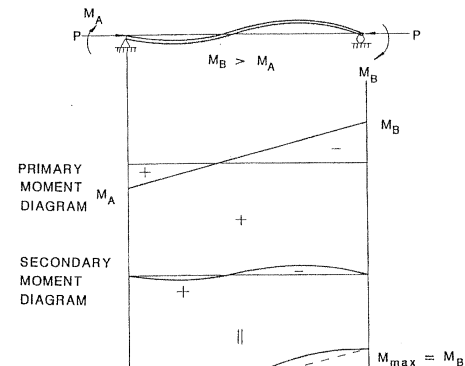
$$y'' = \frac{(M_A \cos kL + M_B)}{EI \sin kL} \sin kx - \frac{M_A}{EI} \cos kx \quad (3.4.10)$$

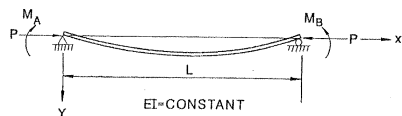
(Note: For  $0 \leq k\bar{x} \leq kL = \pi\sqrt{P/P_e} \leq \pi$ , we have  $\sin k\bar{x} \geq 0$  and  $\cos k\bar{x} \leq 0$ )

The maximum moment is obtained by using the above expressions for  $\sin k\bar{x}$  and  $\cos k\bar{x}$  in the expression  $M = -EIy''$  where  $y''$  is given in Eq. (3.4.10). Thus

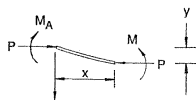
$$\begin{aligned}
 M_{\max} &= \frac{-(M_A \cos kL + M_B)^2}{\sin kL \sqrt{M_A^2 + 2M_A M_B \cos kL + M_B^2}} \\
 &\quad - \frac{M_A^2 \sin kL}{\sqrt{M_A^2 + 2M_A M_B \cos kL + M_B^2}} \\
 &= -\frac{\sqrt{M_A^2 + 2M_A M_B \cos kL + M_B^2}}{\sin kL} \\
 &= -M_B \left[ \frac{(M_A/M_B)^2 + 2(M_A/M_B) \cos kL + 1}{\sin^2 kL} \right] \quad (3.4.15)
 \end{aligned}$$

### 3.4 Beam-Columns Subjected to End Moments





(a)



### 3.4 Beam-Columns Subjected to End Moments

Equations (3.4.15) and (3.4.16) can be written in a combined form as

$$M_{\max} = |M_B| \left[ \sqrt{\frac{(M_A/M_B)^2 + 2(M_A/M_B) \cos kL + 1}{\sin^2 kL}} \right] \quad (3.4.18)$$

where  $M_A/M_B$  is *positive* if the member is bent on *double curvature* and is *negative* if it is bent in *single curvature*. The absolute value for  $M_B$  is used in the coefficient of Eq. (3.4.18) because we are interested only in the magnitude, not the direction of  $M_{\max}$ .

A special case for a beam column bent in single curvature is the case in which the end moments are equal and opposite, i.e., if  $M_A = -M_B = M$ , as shown in Fig. 3.9. For this case the maximum moment is given by substituting  $M_A/M_B = -1$  in Eq. (3.4.18)

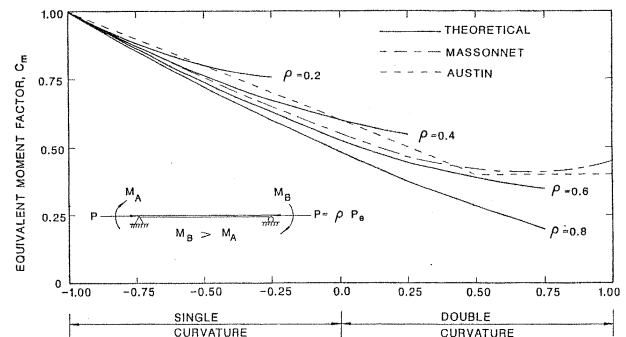
$$M_{\max} = M \sqrt{\frac{2(1 - \cos kL)}{\sin^2 kL}} \quad (3.4.19)$$

### 3.4.2 Concept of Equivalent Moment

Equation (3.4.18) is the expression for the maximum moment for the general case of a beam-column subjected to unequal end moments. The maximum moment may occur at a member's end, and be equal to the larger of the two end moments, or it may occur somewhere within the member whose magnitude is given by Eq. (3.4.18) and whose location is determined by Eq. (3.4.12) or Eq. (3.4.17). For the purpose of design, one needs to know whether the maximum moment occurs at the end or away from the ends, and also the location of the maximum moment if it should occur away from the ends. To eliminate these calculations, the *concept of equivalent moment* is introduced in design practice.

The concept of equivalent moment is shown schematically in Fig. 3.10. The end moments  $M_A$  and  $M_B$  that act on the member are replaced by a pair of equal and opposite equivalent moment  $M_{eq}$ . The magnitude of the equivalent moment is such that the maximum moment produced by it will

### 3.4 Beam-Columns Subjected to End Moments



2. Calculate the equivalent moment by multiplying the  $C_m$  factor by the larger of the two end moments (Eq. 3.4.21).
3. Calculate the maximum moment from Eq. (3.4.19) with  $M = M_{eq}$ .

Equation (3.4.19) can be further simplified as follows. Using the trigonometric identities

$$1 - \cos kL = 2 \sin^2 \frac{kL}{2} \quad (3.4.24)$$

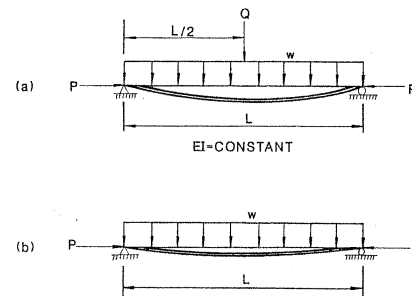
and

$$\sin^2 kL = 4 \sin^2 \frac{kL}{2} \cos^2 \frac{kL}{2} \quad (3.4.25)$$

Eq. (3.4.19) can be written as

$$M_{\max} = M_{eq} \sec \frac{kL}{2} \approx M_{eq} \left( \frac{1}{\cos \frac{kL}{2}} \right) \quad (3.4.26)$$

### 3.5 Superposition of Solutions



example there is a simpler approach. Knowing that the maximum moment occurs at midspan for both the uniformly loaded case (Fig. 3.12b) and the concentrated loaded case (Fig. 3.12c), we can just add the maximum moment for these two cases together to obtain the maximum moment of the combined loading case (Fig. 3.12a). Thus

$$M_{\max} = \text{Eq. (3.2.36)} + \text{Eq. (3.3.17)} \\ = \frac{wL^2}{8} \left[ \frac{2(\sec u - 1)}{u^2} \right] + \frac{QL}{4} \left[ \frac{\tan u}{u} \right] \quad (3.5.2)$$

### 3.5.2 Fixed-Ended Beam-Columns

Another application of the principle of superposition is to determine the fixed-end moments of a beam-column.

#### Uniform Load Case (Fig. 3.13)

### 3.5 Superposition of Solutions

will determine the fixed-end moments ( $M_{FA}$ ,  $M_{FB}$ ) of this member using the principle of superposition. The beam-column in Fig. 3.13a can be decomposed into that of Fig. 3.13b and Fig. 3.13c. To satisfy the continuity condition of zero slopes at the built-in ends, the algebraic sum of the rotation at the ends produced by the uniform load  $w$  (Fig. 3.13b) and that produced by the end moments (Fig. 3.13c) must be zero. Because of symmetry, we need only consider half of the member. By taking the derivative of Eq. (3.2.28) and evaluating the resulting equation at  $x = 0$ , we obtain the A end rotation produced by the uniform load  $w$  as

$$y'_w(0) = \frac{wL^3}{24EI} \left[ \frac{3(\tan u - u)}{u^3} \right] \quad (3.5.3)$$

and from Eq. (3.4.9), by setting  $M_A = M_{FA}$ ,  $M_B = M_{FB} = -M_{FA}$  and  $x = 0$ , we obtain the A end rotation produced by the end moments  $M_{FA}$  and  $M_{FB}$  as

**Concentrated Load Case (Fig. 3.14)**

The same procedure can be used to evaluate the fixed-end moment of a beam-column subjected to a concentrated lateral load  $Q$  acting at midspan of the member (Fig. 3.14a).

Again, because of symmetry, we only need consider half of the member. From Eq. (3.3.14a), by setting  $a = L/2$ , we obtain the A end rotation due to the lateral load  $Q$  as

$$\begin{aligned} y'_O(0) &= \frac{Q}{Elk^2} \frac{\sin \frac{kL}{2}}{\sin kL} - \frac{Q}{2Elk^2} \\ &= \frac{QL^2}{8Elu^2} [\sec u - 1] \end{aligned} \quad (3.5.8)$$

where  $u = kL/2$ .

The end rotation at the A end due to the end moments  $M$

**3.6 Basic Differential Equations**

and  $M_{FB} = -M_{FA}$  is obtained from Eq. (3.5.5)

$$y'_M(0) = \frac{M_{FA}L}{2EI} \left[ \frac{\tan u}{u} \right] \quad (3.5.5)$$

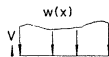
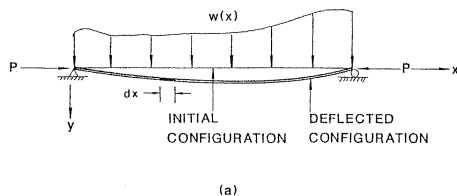
Since the slope at the fixed-end must be zero, we must satisfy the condition

$$y'_O(0) + y'_M(0) = 0 \quad (3.5.9)$$

Upon substituting Eqs. (3.5.8) and (3.5.5) into (3.5.9), we can determine the fixed-end moment as

$$M_{FA} = \frac{-QL}{8} \left[ \frac{2(\sec u - 1)}{u \tan u} \right] = \frac{-QL}{8} \left[ \frac{2(1 - \cos u)}{u \sin u} \right] \quad (3.5.10)$$

The minus sign indicates that the direction of  $M_{FA}$  is opposite to that shown in Fig. 3.14. The term  $QL/8$  that appeared in Eq. (3.5.10) represents the fixed-end moment of the member when  $P$  is absent. The



## 3.6 Basic Differential Equations

or, neglecting the higher order terms involving  $(dx)^2$ , we obtain

$$\frac{dM}{dx} - V - P \frac{dy}{dx} = 0 \quad \text{or} \quad \frac{dM}{dx} = V + P \frac{dy}{dx} \quad (3.6.3)$$

Equation (3.6.1) states that there is no change in longitudinal force in the segment. That is to say, the force  $P$  remains constant in the element. Equation (3.6.2) states that the rate of change in transverse force across the segment is equal numerically to the magnitude of the applied transverse load  $w(x)$ , which is assumed to be constant along the infinitesimal length  $dx$ . Equation (3.6.3) states that the rate of change in moment across the segment is equal numerically to the transverse force  $V$  plus the longitudinal force  $P$  times the change in slope  $dy/dx$  of the infinitesimal element.

For small deflection, we can write  $M = -EI d^2y/dx^2$ . Upon substitution into Eq. (3.6.3) and rearranging, we obtain

### 3.6.1 Fixed-Fixed Beam-Column with Concentrated Load at Midspan

The beam-column under investigation is shown in Fig. 3.16. Because of symmetry, only half of the member is considered in the analysis. Since the transverse force acting on any cut section is constant and is equal to  $Q/2$ , we have, from Eq. (3.6.5),

$$\frac{d^3y}{dx^3} + k^2 \frac{dy}{dx} = -\frac{Q}{2EI} \quad (3.6.10)$$

From Eq. (3.6.8), the general solution is

$$y = A \sin kx + B \cos kx + C - \frac{Q}{2EI k^2} x \quad (3.6.11)$$

The constants  $A$ ,  $B$ , and  $C$  are determined from the boundary conditions

### 3.6 Basic Differential Equations

Using these boundary conditions, we find

$$A = \frac{Q}{2EI k^3} \quad (3.6.13)$$

$$B = \frac{Q}{2EI k^3 \sin \frac{kL}{2}} \left( \cos \frac{kL}{2} - 1 \right) \quad (3.6.14)$$

$$C = \frac{Q}{2EI k^3 \sin \frac{kL}{2}} \left( 1 - \cos \frac{kL}{2} \right) \quad (3.6.15)$$

With these constants, the deflection function (3.6.11) is fully determined

$$y = \frac{Q}{2EI k^3 \sin \frac{kL}{2}} \left[ \sin kx + \left( \cos \frac{kL}{2} - 1 \right) \cos kx + \left( 1 - \cos \frac{kL}{2} \right) \right]$$

**Table 3.5** Theoretical and Design Moment Amplification Factors for a Fixed-Ended Beam-Column Loaded by a Concentrated Lateral Load at Midspan

| $u = \frac{kL}{2} = \frac{\pi}{2} \sqrt{\frac{P}{P_{ek}}}$ | Theoretical<br>Eq. (3.6.18) | Design<br>Eq. (3.6.19) |
|------------------------------------------------------------|-----------------------------|------------------------|
| 0                                                          | 1.000                       | 1.000                  |
| 0.20                                                       | 1.003                       | 1.003                  |
| 0.40                                                       | 1.014                       | 1.013                  |
| 0.60                                                       | 1.031                       | 1.030                  |
| 0.80                                                       | 1.057                       | 1.055                  |
| 1.00                                                       | 1.093                       | 1.090                  |
| 1.20                                                       | 1.140                       | 1.137                  |
| 1.40                                                       | 1.203                       | 1.198                  |
| $\pi/2$                                                    | $\infty$                    | $\infty$               |

### 3.6 Basic Differential Equations

conditions

$$y(0) = 0, \quad y'(0) = 0, \quad y(L) = 0, \quad y'(L) = 0 \quad (3.6.23)$$

or, alternatively, by making use of the condition of symmetry, the four boundary conditions are

$$y(0) = 0, \quad y'(0) = 0, \quad y'\left(\frac{L}{2}\right) = 0, \quad y''\left(\frac{L}{2}\right) = 0 \quad (3.6.24)$$

Using either set of these four conditions, we find

$$A = \frac{wL}{2EI k^3} \quad (3.6.25)$$

$$B = \frac{wL}{2EI k^3 \tan \frac{kL}{2}} \quad (3.6.26)$$